Lab Observation

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12. Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux).

Aim:

To demonstrate virtualization by installing and using a Type-2 Hypervisor (VMware Workstation) on a host operating system and creating and configuring a Virtual Machine (VM).

Procedure:

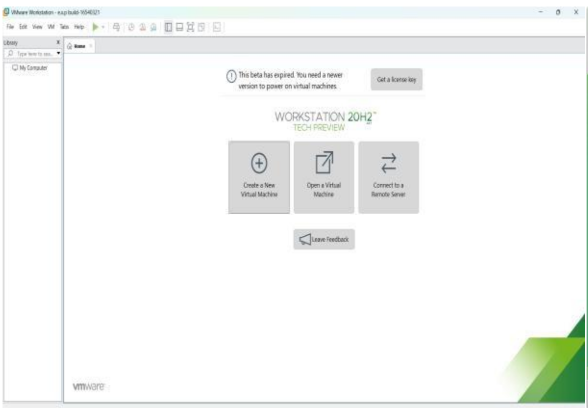
Step 1 – Start VMware Workstation

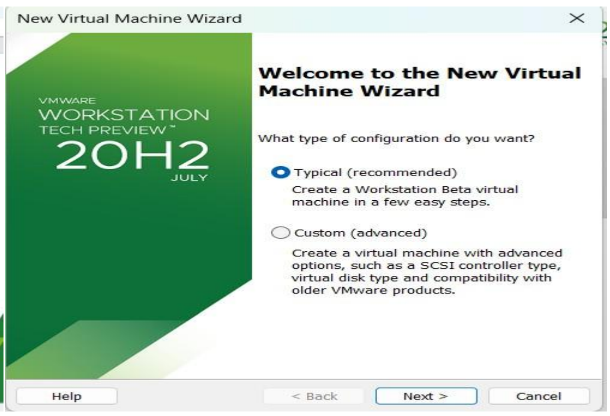
1. Open VMware Workstation in Windows.
2. VMware Workstation acts as the Type-2 Hypervisor.
3. Click Create a New Virtual Machine.

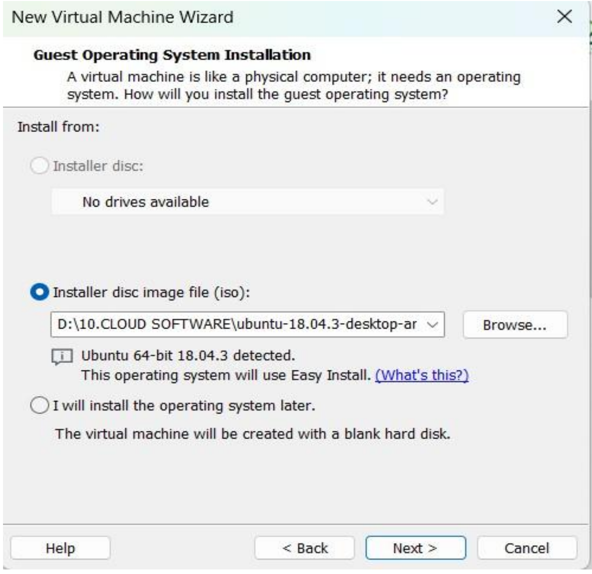
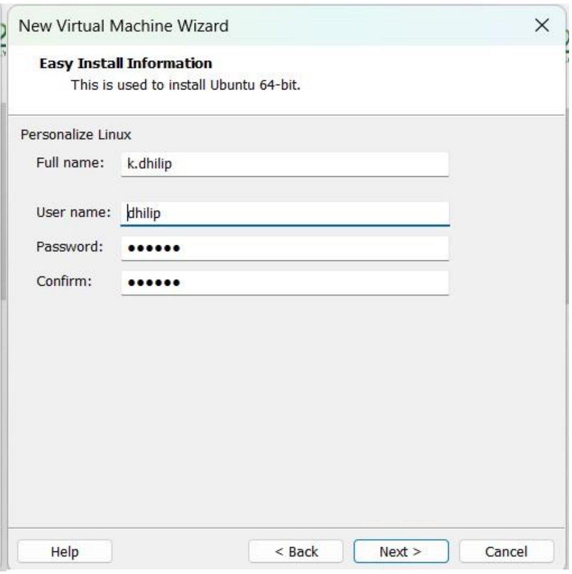
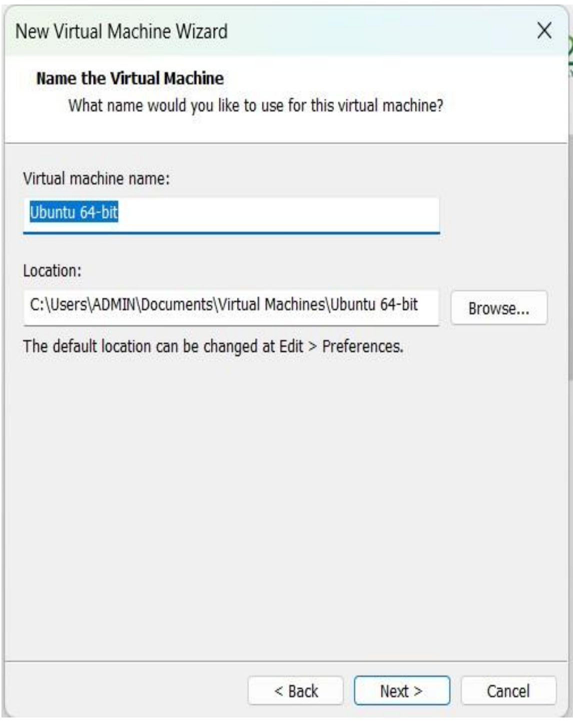
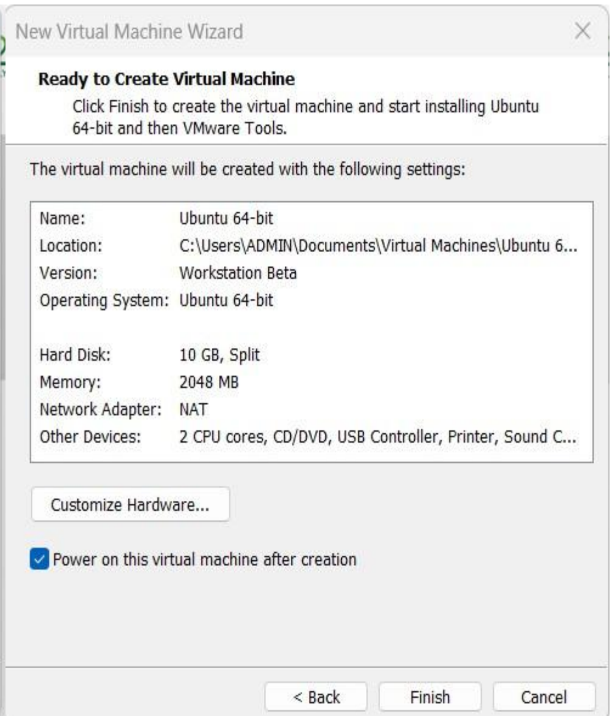
Step 2 – Create the VM

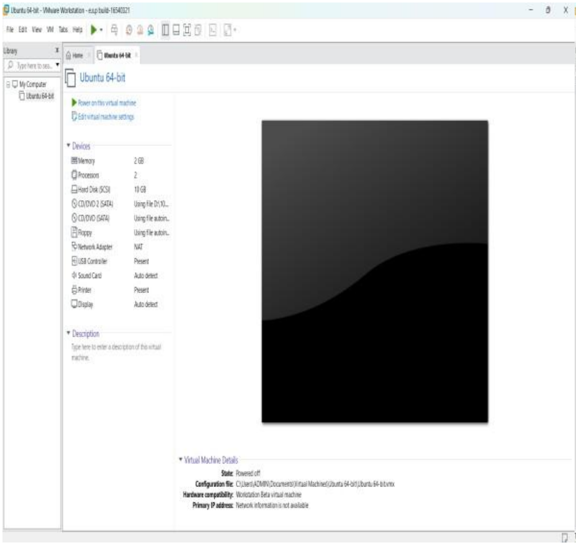
1. Select Typical (Recommended).
2. Click Next.
3. Select the Ubuntu ISO file.
4. Click Next.
5. Open Terminal and Run Comments.

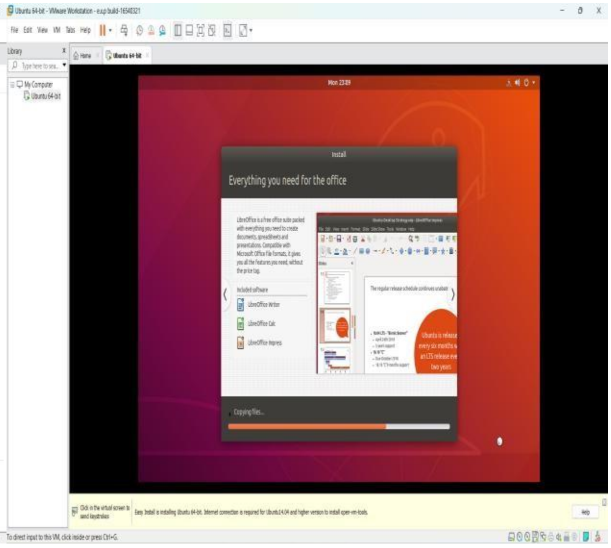
Implementation:

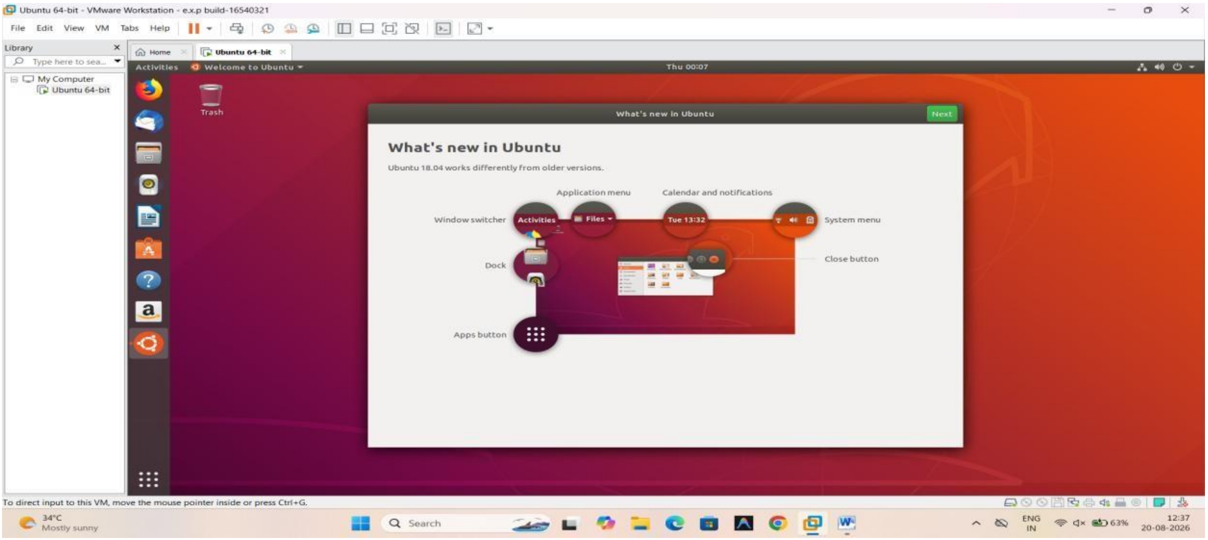


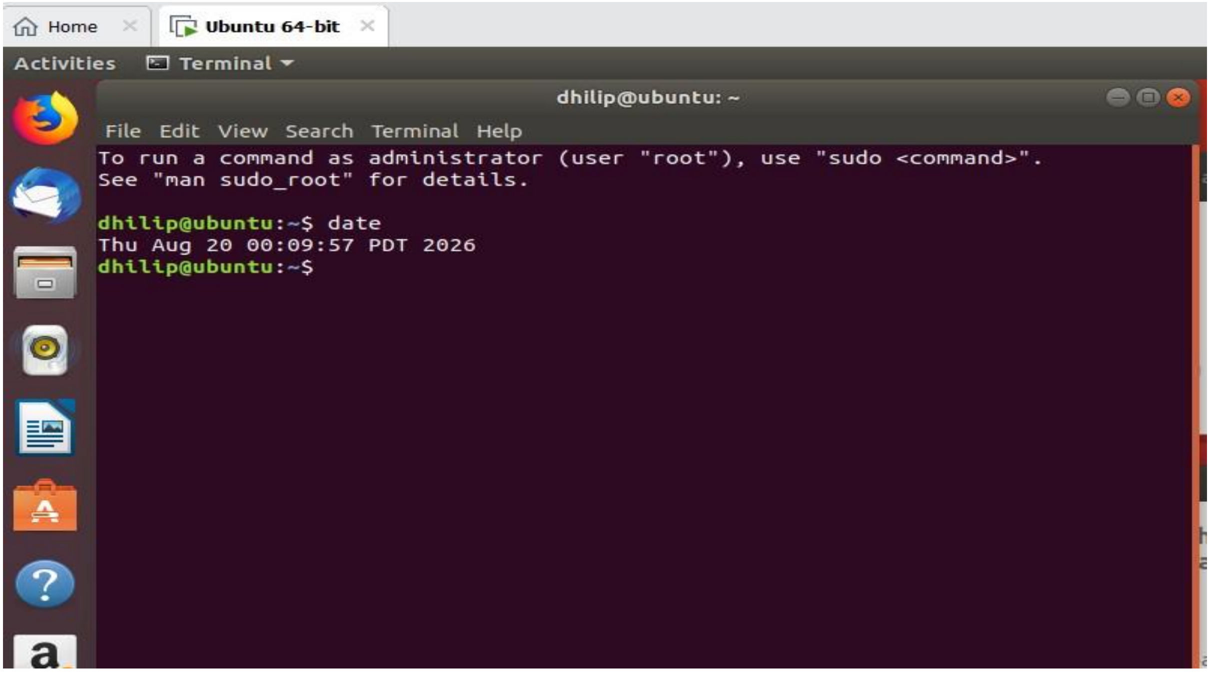



New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
D:\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-ar
Browse...
Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel

New Virtual Machine Wizard
Easy Install Information
This is used to install Ubuntu 64-bit.
Personalize Linux
Full name: k.dhilip
User name: dhilip
Password: ••••••
Confirm: ••••••
Help
< Back
Next >
Cancel

New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit
Location:
C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu 64-bit
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel

New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Ubuntu 64-bit
Location: C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu 6...
Version: Workstation Beta
Operating System: Ubuntu 64-bit
Hard Disk: 10 GB, Split
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel






Ubuntu 64-bit - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Home
Ubuntu 64-bit
My Computer
Activities
Welcome to Ubuntu
Thu 00:07
Trash
What's new in Ubuntu
Next
What's new in Ubuntu
Ubuntu 18.04 works differently from older versions.
Application menu
Calendar and notifications
Window switcher
System menu
Dock
Close button
Apps button
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
34°C
Mostly sunny
Search
ENG
IN
12:37
20-08-2026


Home
Ubuntu 64-bit
Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ date
Thu Aug 20 00:09:57 PDT 2026
dhilip@ubuntu:~$

13. **Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.**

Aim:

To create a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage disk using a Type 2 virtualization software (VMware Workstation).

Procedure:

Step 1: Open VMware

1. Open VMware Workstation.
2. Click Create a New Virtual Machine.
3. Select Typical (Recommended).
4. Click Next.

Step 2: Select Operating System

1. Select Installer disc image file (ISO).
2. Click Browse.
3. Select your Ubuntu ISO file.
4. Click Next.

Step 3: Enter VM Information Enter:

- Full Name: Your name
- Username: Your preferred username
- Password: Your preferred password

Click Next.

If VMware does not show this screen, you can install Ubuntu normally after creating the VM.

Step 4: Give VM Name and Location

For example:

- Virtual Machine Name: Ubuntu-15GB
- Location: Choose where you want to store the VM.

Click Next.

Step 5: Set Storage

When VMware asks for disk capacity:

- Maximum disk size: 15 GB

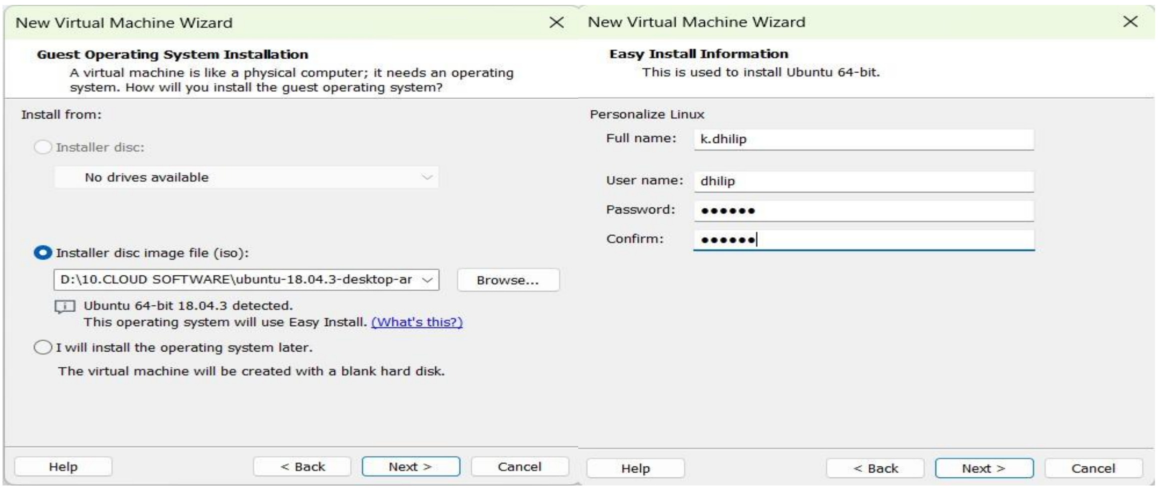
- Select Store virtual disk as a single file.

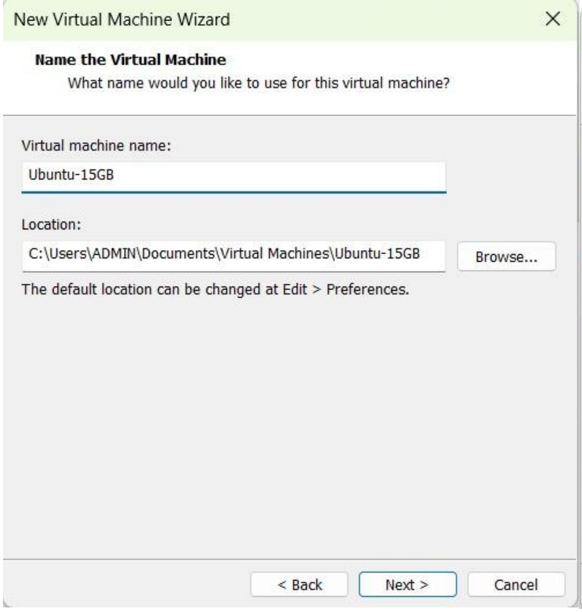
Click Next.

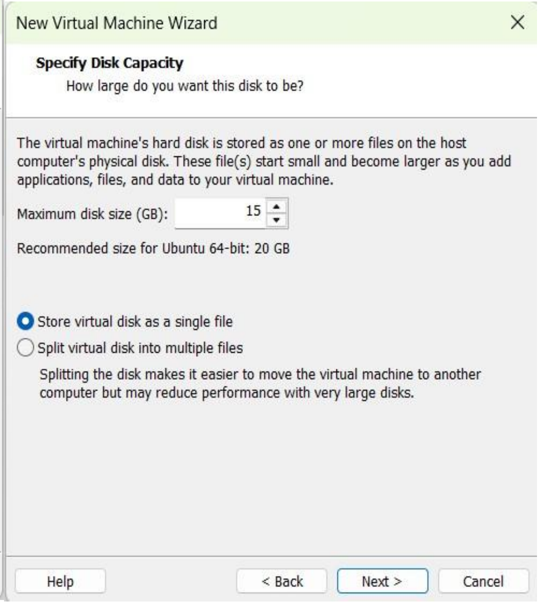
Step 6: Set CPU and RAM

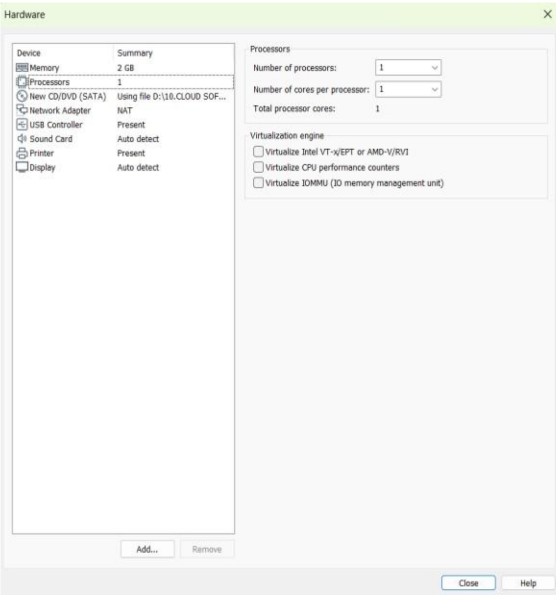
Before finishing, click Customize Hardware.

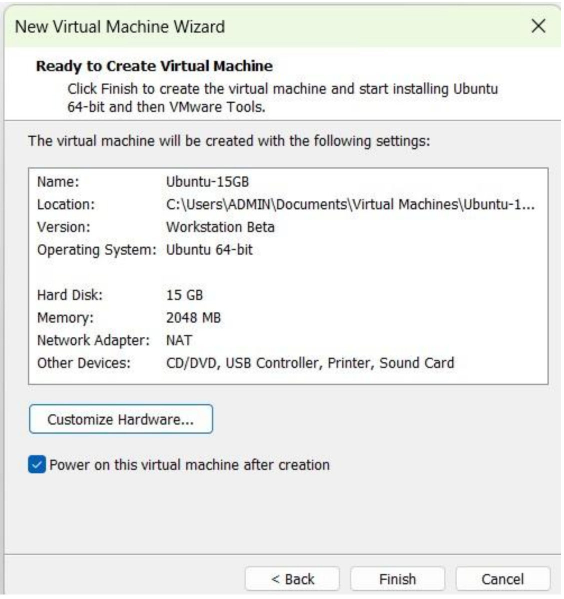
Implementation:

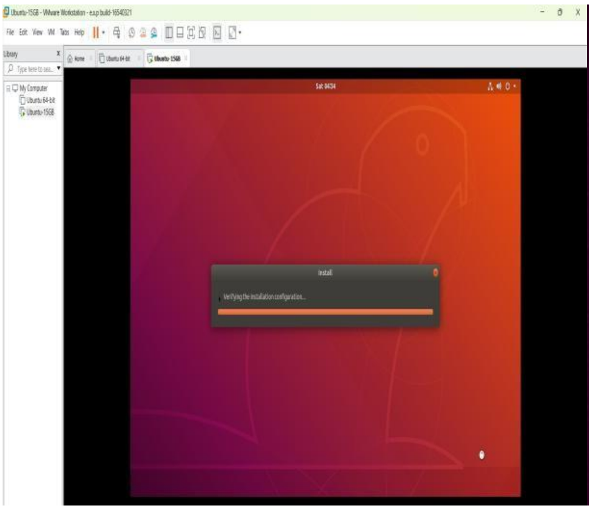


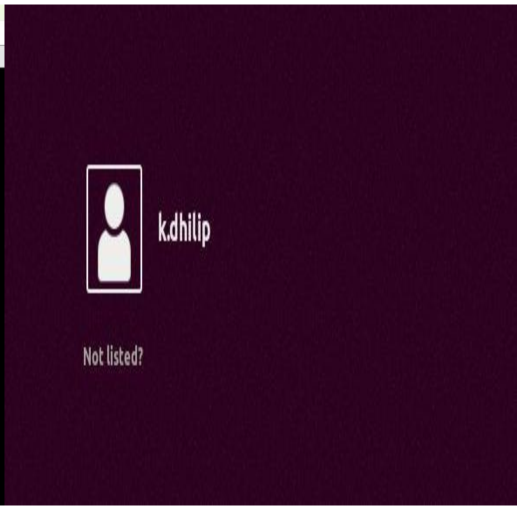

New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu-15GB
Location:
C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-15GB
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel

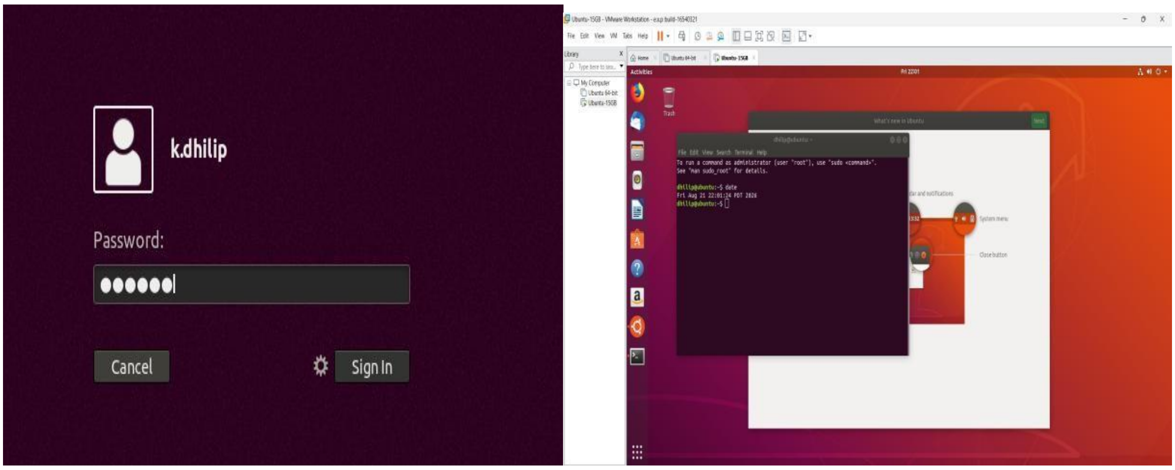

New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 15
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel


Hardware
Device
Summary
Memory
2 GB
Processors
1
New CD/DVD (SATA)
Using file D:\10.CLOUD SOF...
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Printer
Present
Display
Auto detect
Processors
Number of processors:
1
Number of cores per processor:
1
Total processor cores:
1
Virtualization engine
Virtualize Intel VT-x/EPT or AMD-V/RVI
Virtualize CPU performance counters
Virtualize IOMMU (IO memory management unit)
Add...
Remove
Close
Help


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Ubuntu-15GB
Location: C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-1...
Version: Workstation Beta
Operating System: Ubuntu 64-bit
Hard Disk: 15 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: CD/DVD, USB Controller, Printer, Sound Card
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel




k.dhilip
Not listed?



14. **Create a Virtual Hard Disk and allocate the storage using VM ware Workstation**

Aim:

To create a Virtual Hard Disk and allocate storage to it using VMware Workstation.

Procedure:

1. Open VMware Workstation.
2. Select your existing Ubuntu VM.
3. Make sure the VM is powered off.
4. Right-click the VM and select Settings.
5. Click Add... at the bottom of the Hardware tab.
6. Select Hard Disk.
7. Click Next.
8. Select the disk type:
   SCSI (recommended/default).
9. Click Next.

10. Select:
    Create a new virtual disk.
11. Click Next.
12. Enter the required disk size.

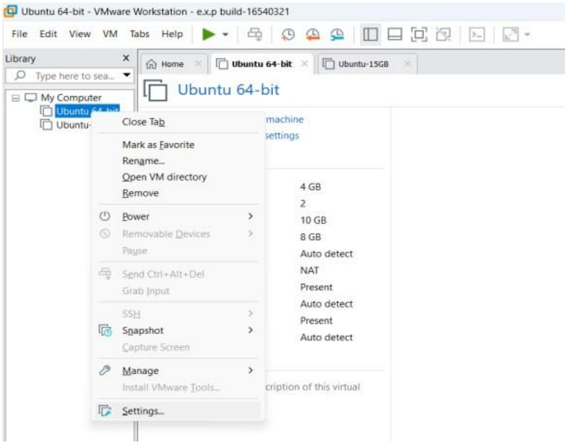
13. Select:

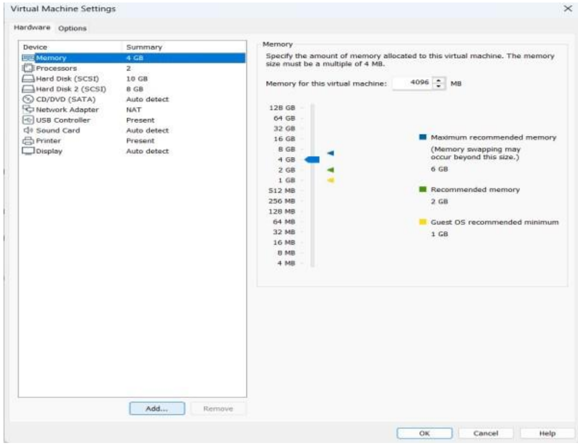
14. Click Next.
15. Give the virtual disk a name, for example:

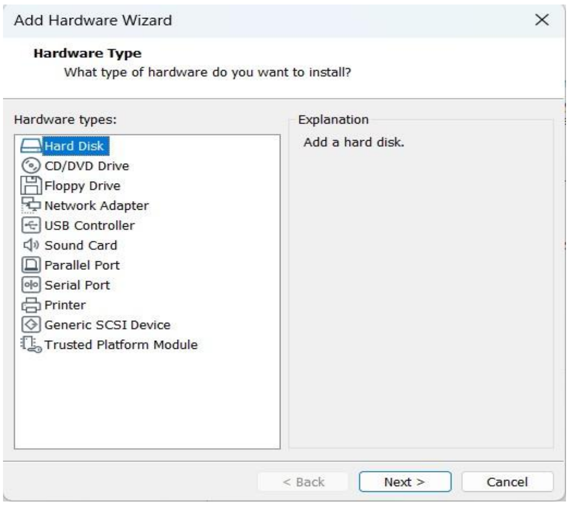
16. Click Finish.
17. Click OK to save the VM settings.

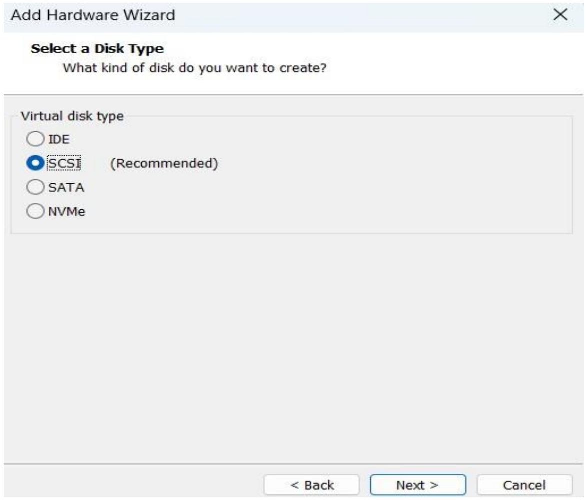
Start the Ubuntu VM and open Terminal and run the comments.

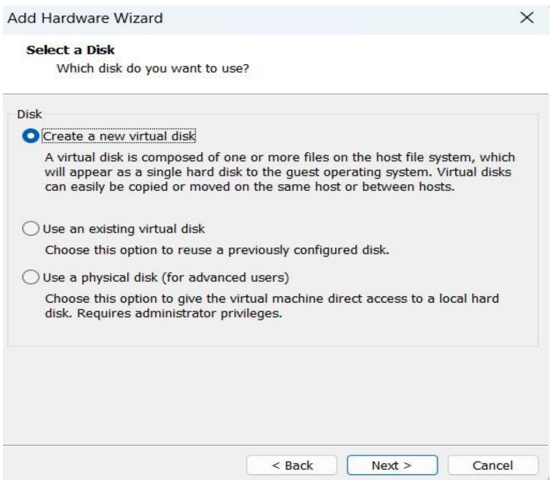
Implementation:

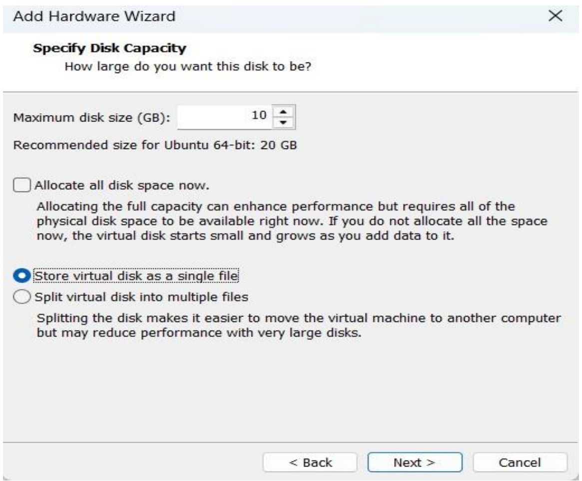


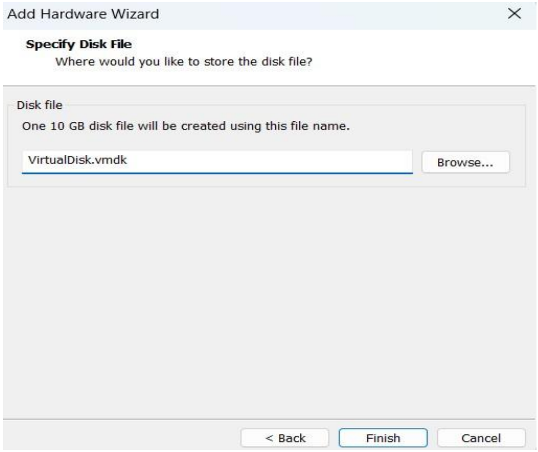


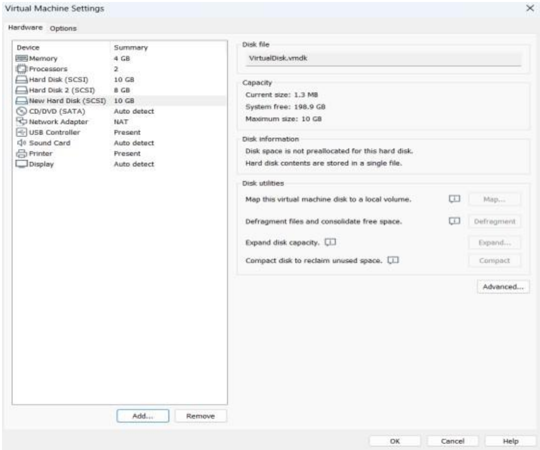


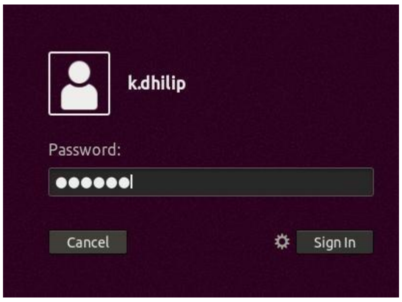


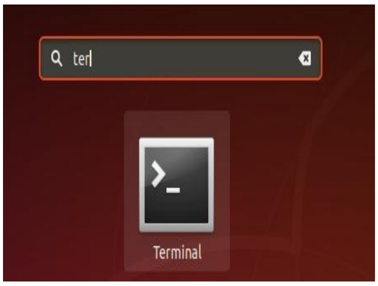




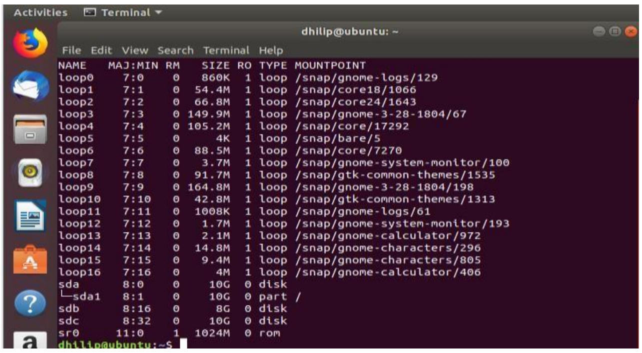








Commend: lsblk



15. **Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM**

Aim:

To create a snapshot of a Virtual Machine using VMware Workstation and test it by restoring the VM to the previous saved state.

Procedure:

Step 1: Start the VM

1. Open VMware Workstation.
2. Select your Ubuntu VM.
3. Start the VM.

4. Wait until Ubuntu loads completely.

Step 2: Create a Snapshot

1. In VMware Workstation, select the running Ubuntu VM.
2. Go to:

VM → Snapshot → Take Snapshot

3. Enter a name:

Before_Test

4. You can enter a description such as:

Snapshot before making changes

5. Click Take Snapshot.

The current state of the VM is now saved.

Step 3: Make a Change

Open Ubuntu Terminal and create a test file:

touch test_snapshot.txt

Check it:

ls

You should see: test_snapshot.txt

Step 4: Restore the Previous

Snapshot

Now restore the VM to the snapshot you created earlier.

1. In VMware Workstation, go to:

VM → Snapshot → Snapshot Manager

2. Select:

Before_Test

3. Click Go To.

4. Confirm the restore operation if VMware asks.

VMware will return the VM to the state when the snapshot was taken.

Step 5: Test the Snapshot

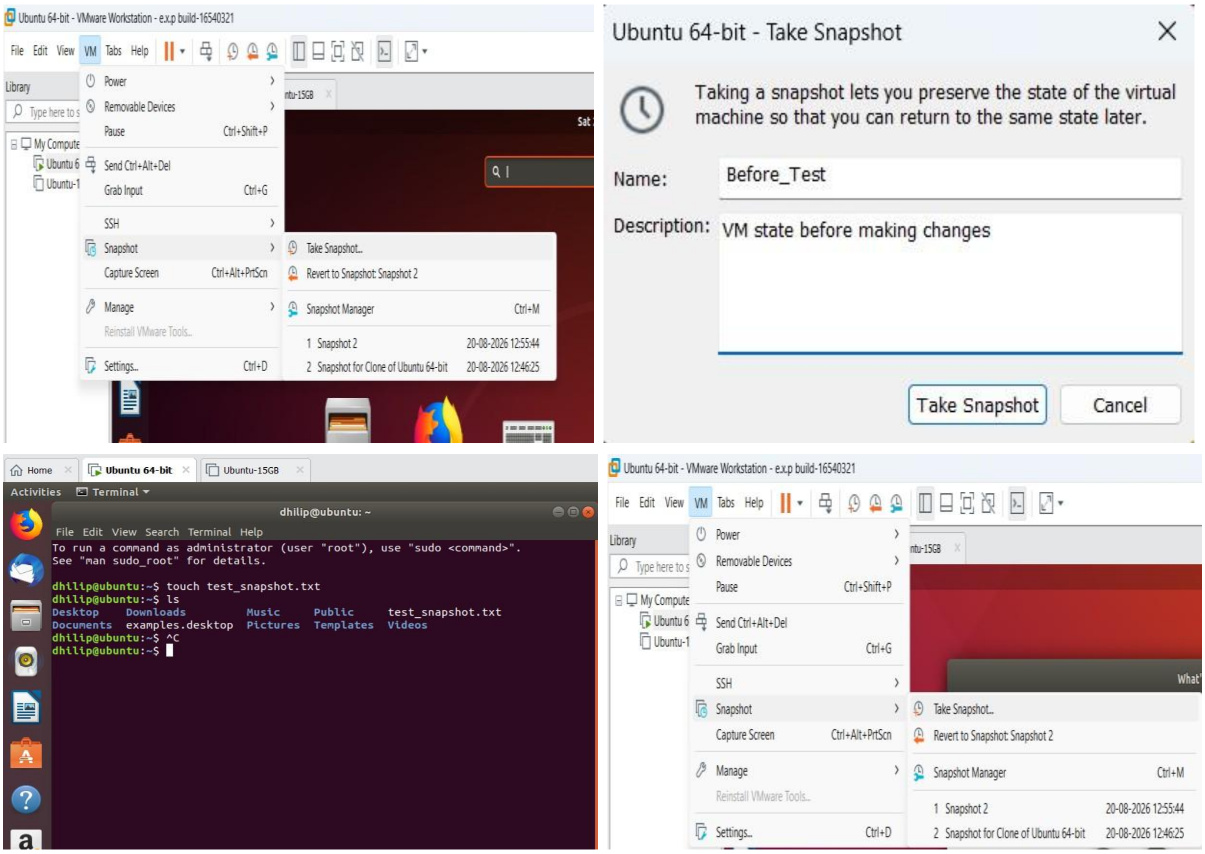
After Ubuntu starts again, open Terminal and run:

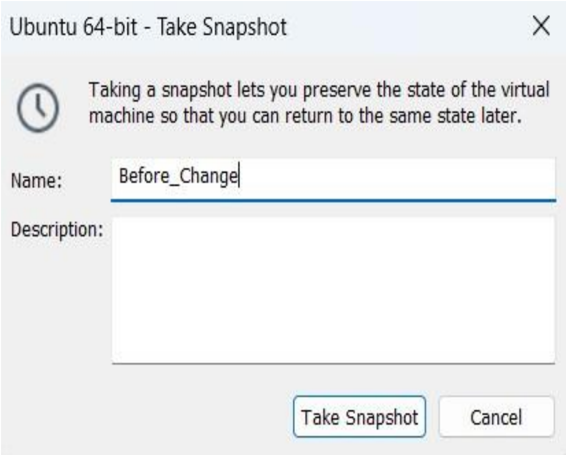
ls

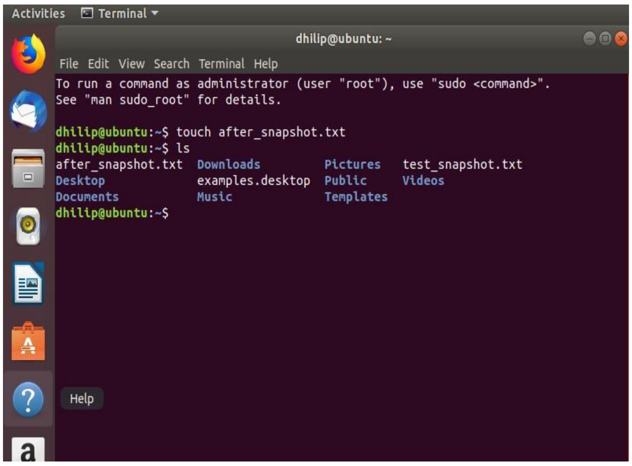
The file:

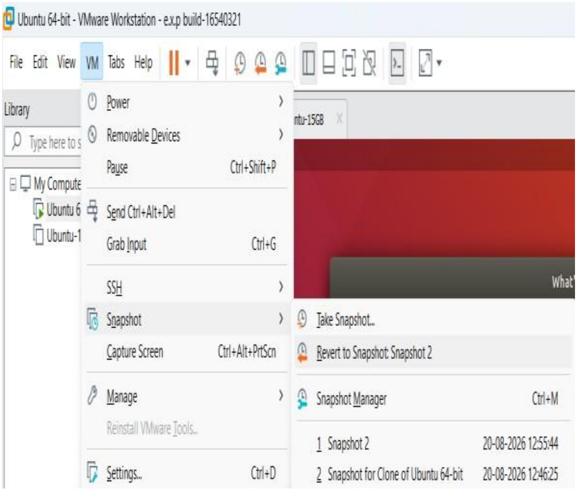
test_snapshot.txt

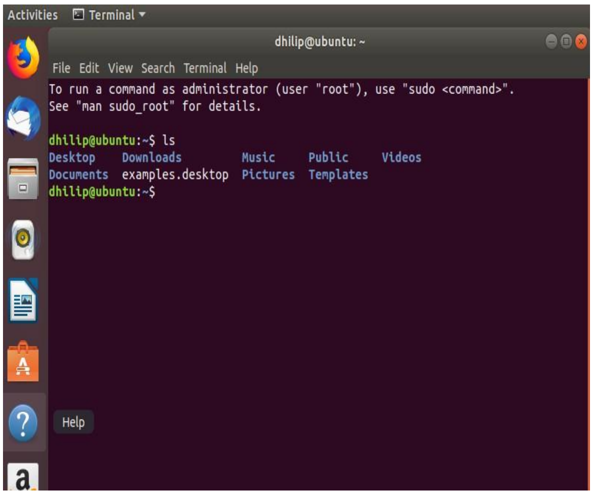
Implementation:











16. **Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM**

Aim:

To create a clone of an existing Virtual Machine using VMware Workstation and test the cloned VM by loading and running it successfully.

Requirements:

- VMware Workstation
- An existing Ubuntu/Linux VM

- Sufficient storage space for the cloned VM

Procedure:

Step 1: Power Off the Original VM

1. Open VMware Workstation.
2. Select your existing Ubuntu VM.
3. Make sure the VM is powered off.
    - If it is running, shut down Ubuntu normally.

Step 2: Create the Clone

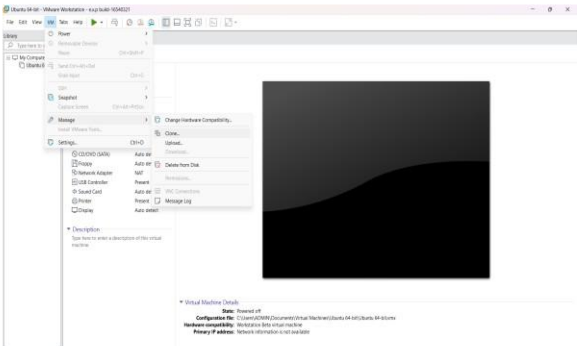
1. Select the original VM.
2. Go to:

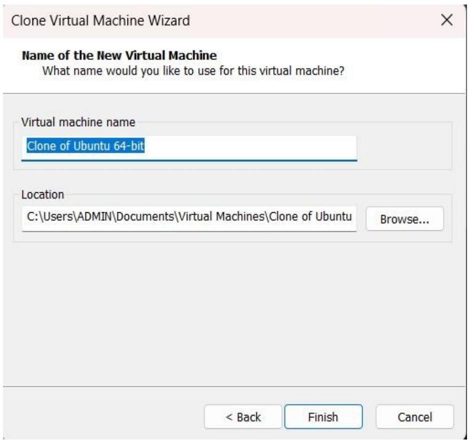
VM → Manage → Clone

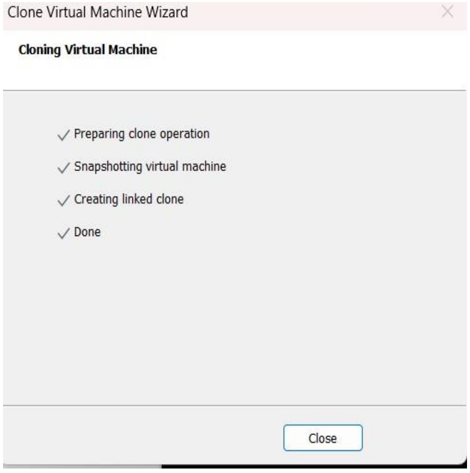
3. The Clone Virtual Machine Wizard will open.
4. Click Next.
5. Select:

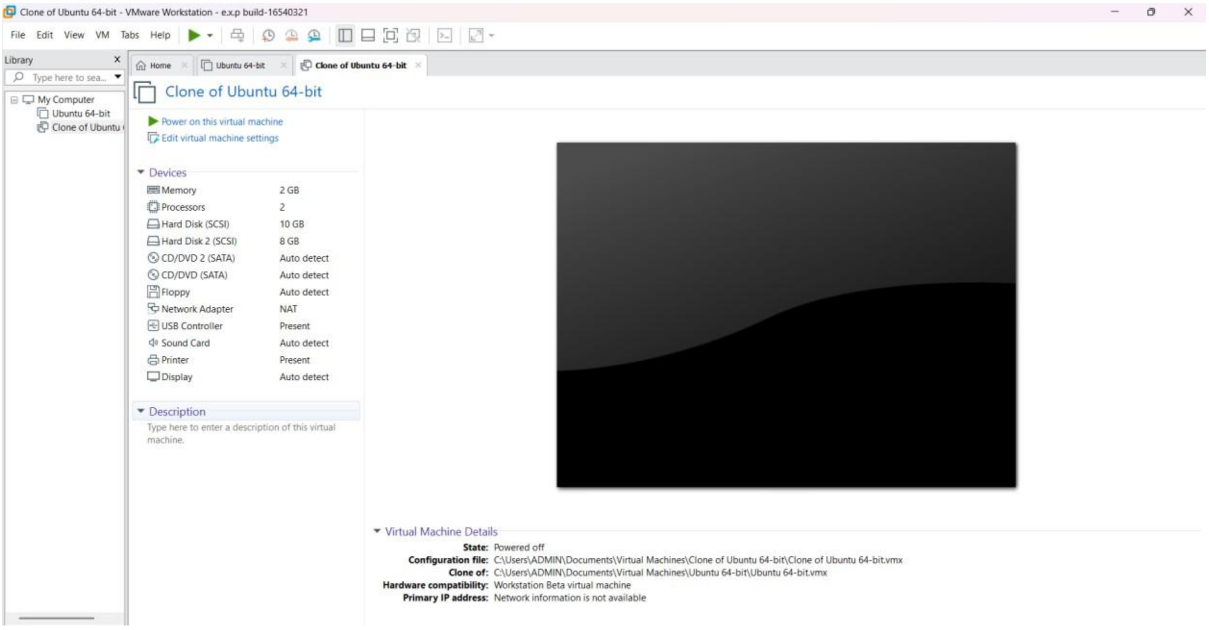
The current state in the virtual machine

6. Click Next.




Clone Virtual Machine Wizard
Name of the New Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name
Clone of Ubuntu 64-bit
Location
C:\Users\ADMIN\Documents\Virtual Machines\Clone of Ubuntu
Browse...
< Back
Finish
Cancel


Clone Virtual Machine Wizard
Cloning Virtual Machine
Preparing clone operation
Snapshotting virtual machine
Creating linked clone
Done
Close


Clone of Ubuntu 64-bit - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Clone of Ubuntu
Home
Ubuntu 64-bit
Clone of Ubuntu 64-bit
Clone of Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 2 GB
Processors 2
Hard Disk (SCSI) 10 GB
Hard Disk 2 (SCSI) 8 GB
CD/DVD 2 (SATA) Auto detect
CD/DVD (SATA) Auto detect
Floppy Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\ADMIN\Documents\Virtual Machines\Clone of Ubuntu 64-bit\Clone of Ubuntu 64-bit.vmx
Clone of: C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation Beta virtual machine
Primary IP address: Network information is not available

17.Change Hardware compatibility of a VM (Either by clone/create new one) which is already created and configured and run simple unix commands

Aim:

To change the hardware compatibility of an already created and configured virtual machine in VMware Workstation and verify the VM by running simple UNIX commands.

Procedure:

Step 1: Open VMware Workstation

1. Open VMware Workstation in Windows.
2. Select your already created Ubuntu VM.
3. Do not start the VM.
4. Make sure the Ubuntu VM is powered off.

Step 2: Open Virtual Machine Settings

1. Select the Ubuntu VM.
2. Click VM from the top menu.
3. Select Manage.
4. Click Change Hardware Compatibility.

Step 3: Select Hardware Compatibility

1. A Change Hardware Compatibility Wizard will open.
2. Click Next.
3. VMware will display the current hardware compatibility version.
4. Select a newer hardware compatibility version available in your VMware Workstation.
5. Click Next.

Example: If your current version is older, select the latest compatible version offered by your VMware Workstation.

Step 4: Choose the Conversion Option

1. VMware may ask whether to:
    - Create a new virtual machine, or
    - Change the existing virtual machine.
2. Select the appropriate option for the existing VM.
3. Click Next.

Step 5: Complete the Hardware Compatibility Change

1. Review the settings.
2. Click Finish.
3. Wait until VMware completes the hardware compatibility change.

☑ The VM's hardware compatibility has now been changed.

Step 6: Start the Ubuntu VM

1. Select the modified Ubuntu VM.
2. Click Power on.
3. Wait for Ubuntu to boot completely.

Step 7: Open Terminal

Open the Ubuntu Terminal using:

Ctrl + Alt + T

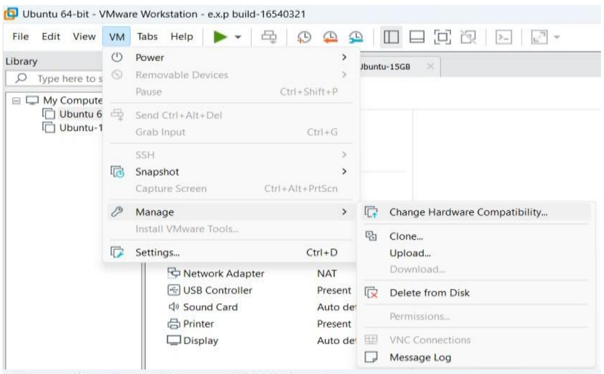
Step 8: Run Simple UNIX Commands Run

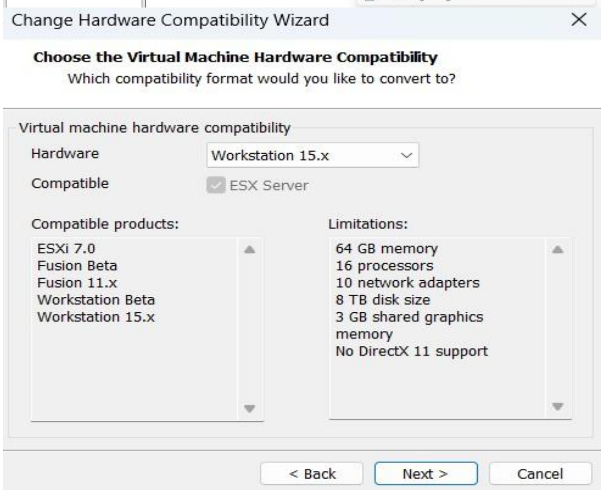
the following commands one by one.

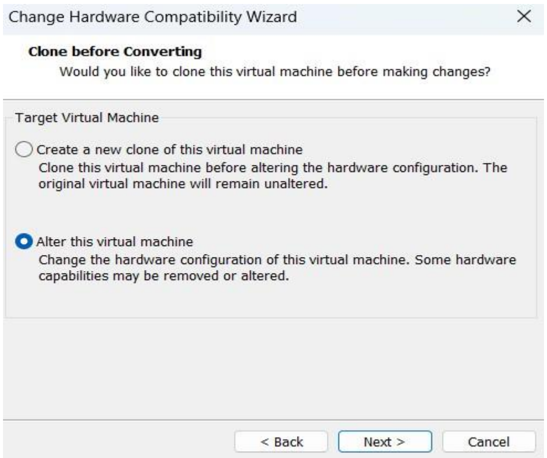
1. Display current user:

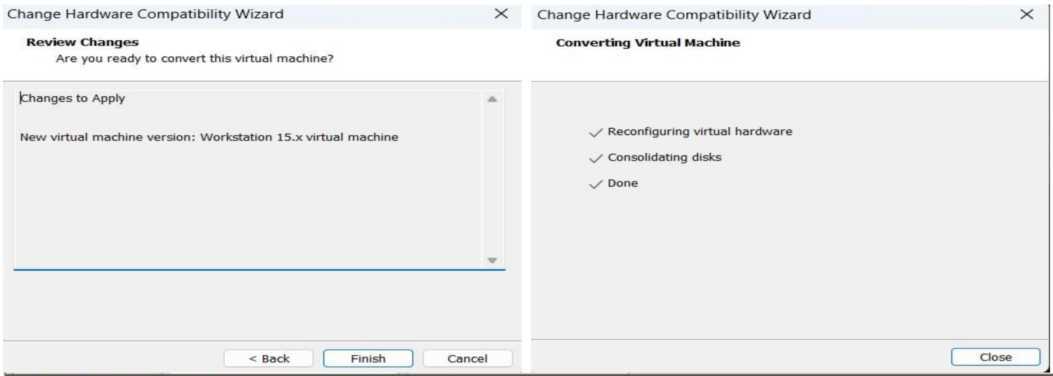
Whoami

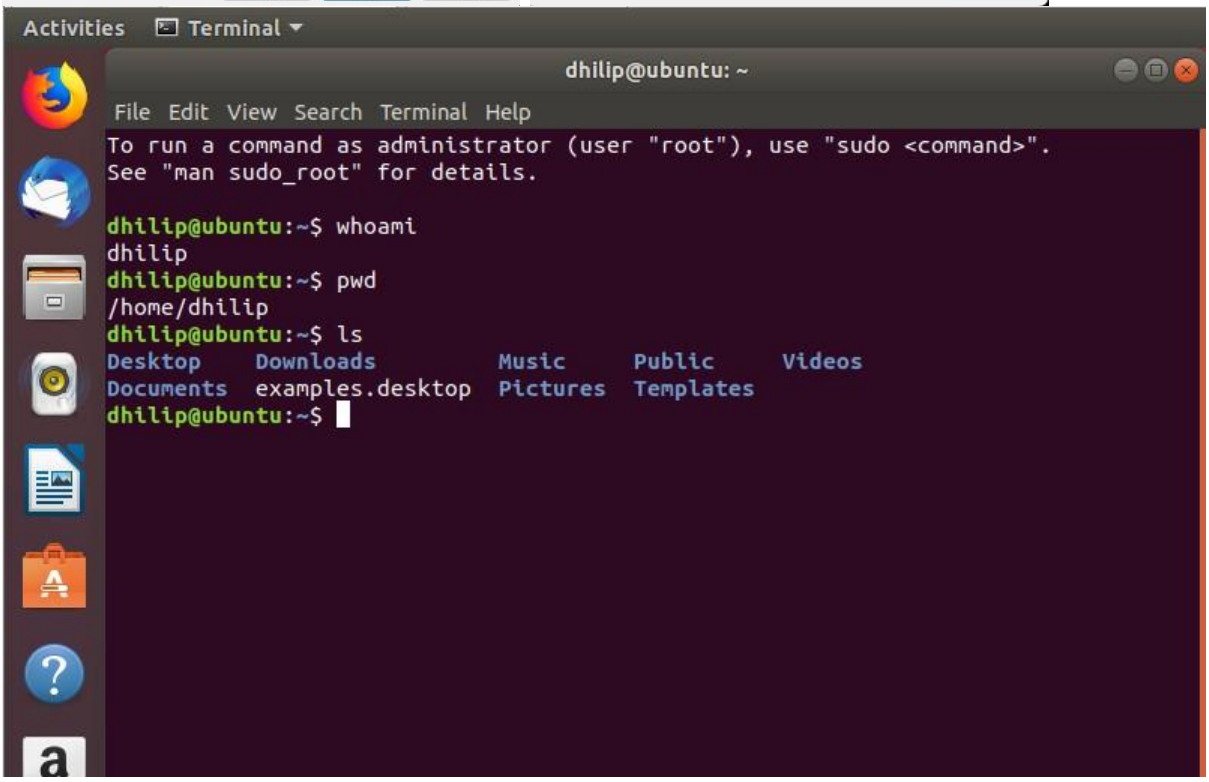
Implementation:

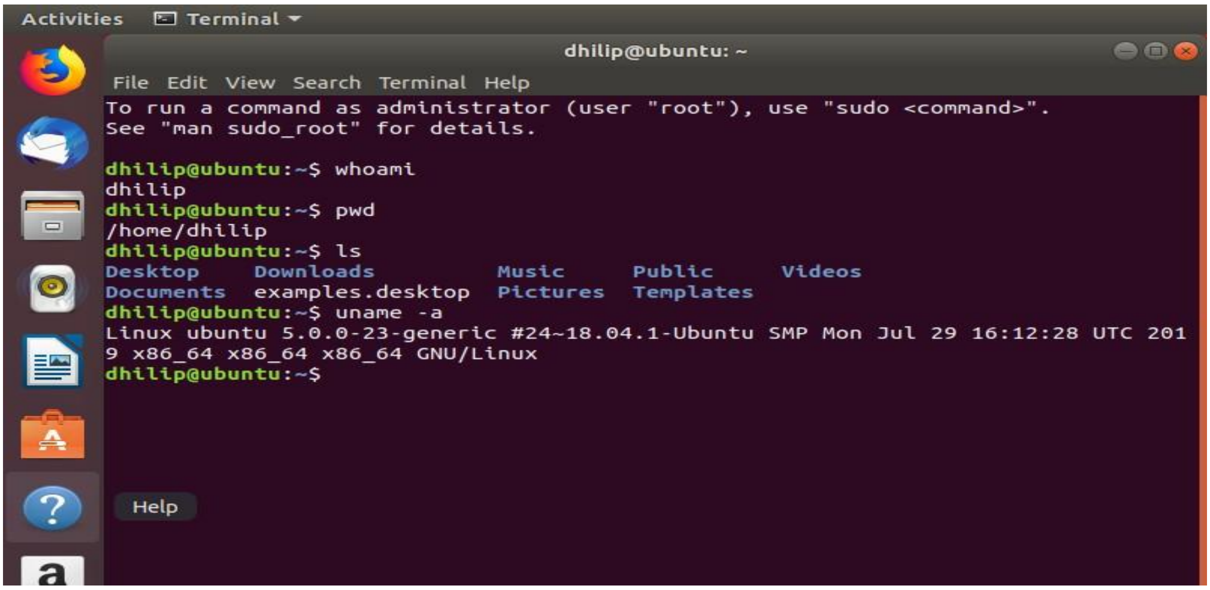







Change Hardware Compatibility Wizard
Review Changes
Are you ready to convert this virtual machine?
Changes to Apply
New virtual machine version: Workstation 15.x virtual machine
< Back
Finish
Cancel
Change Hardware Compatibility Wizard
Converting Virtual Machine
Reconfiguring virtual hardware
Consolidating disks
Done
Close


Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ whoami
dhilip
dhilip@ubuntu:~$ pwd
/home/dhilip
dhilip@ubuntu:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
dhilip@ubuntu:~$


Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ whoami
dhilip
dhilip@ubuntu:~$ pwd
/home/dhilip
dhilip@ubuntu:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$
Help

18. **Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.**

Aim:

To install VMware Workstation and create virtual machines with different operating systems such as Ubuntu Linux on the existing Windows environment.

Procedure:

Step 1: Open VMware Workstation

1. Open VMware Workstation in Windows.
2. Click Create a New Virtual Machine.

Step 2: Select Installation Method

1. Select Typical (recommended).
2. Click Next.

Step 3: Select the Operating System ISO

1. Select Installer disc image file (ISO).
2. Click Browse.
3. Select the Linux/Windows ISO file.
4. Click Next.

Example:

Ubuntu ISO → ubuntu.iso

Step 4: Select Operating System

1. Select the appropriate operating system.
2. For Ubuntu, select:

Linux Ubuntu
64-bit

3. Click Next.

Step 5: Give Virtual Machine Name

Enter a name such as:

Ubuntu-VM

Select a location where you want to store the VM.

Click Next.

Step 6: Configure Disk

1. Specify the disk size.

For example:

20 GB

2. Select Store virtual disk as a single file.
3. Click Next. Step 7: Customize Hardware

Click Customize Hardware.

Configure:

- Memory: 2 GB or more
- Processor: 2 cores
- Network Adapter: NAT

Click Close → Finish.

Step 8: Install the Operating System

1. Start the virtual machine by clicking Power on this virtual machine.
2. The operating system installer will start.
3. Follow the installation instructions.
4. Create a username and password when requested.
5. Wait for the installation to complete. Step 9: Restart the Virtual Machine

After installation:

1. Restart the VM.
2. Remove the ISO if VMware asks you to do so.
3. Ubuntu will boot from the virtual hard disk.

Step 10: Verify the Installation Open

the Terminal in Ubuntu and run:

uname -a

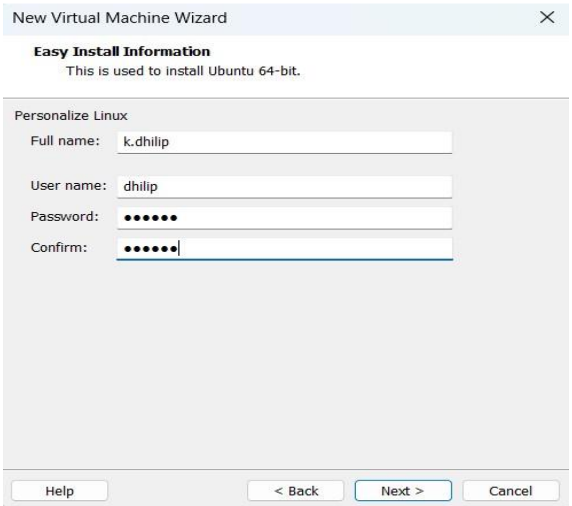
Then:

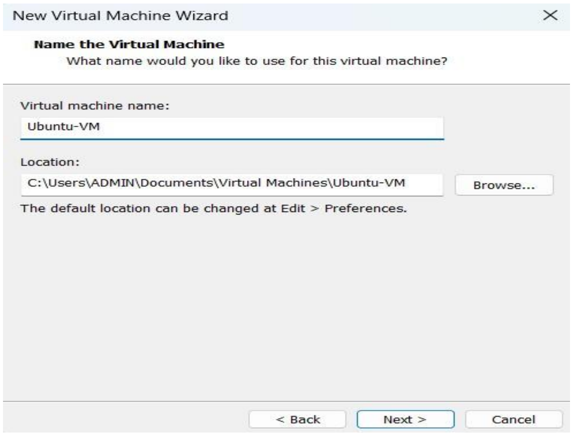
ls

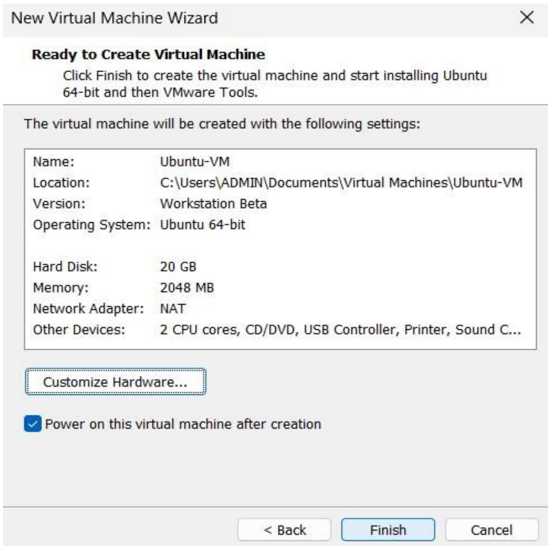
And

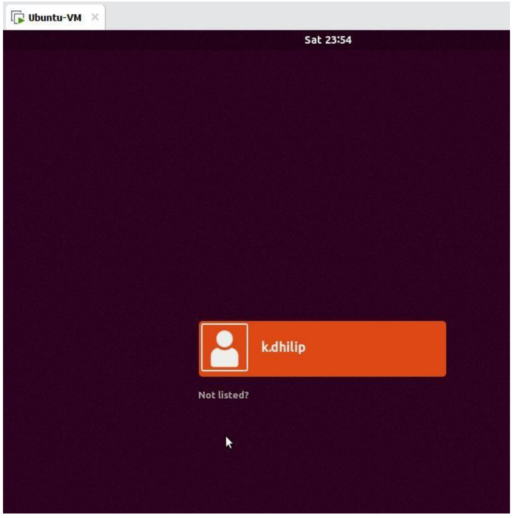
Pwd

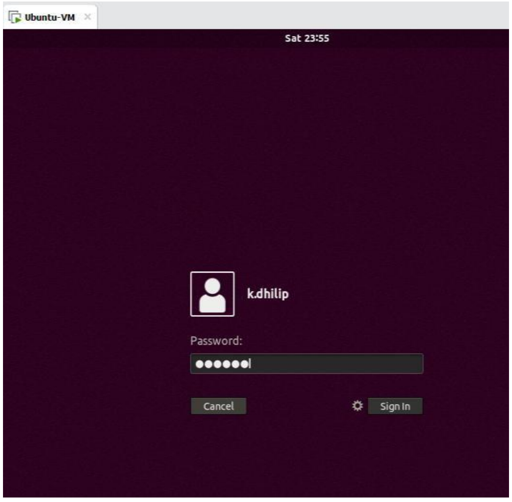
Implementation:

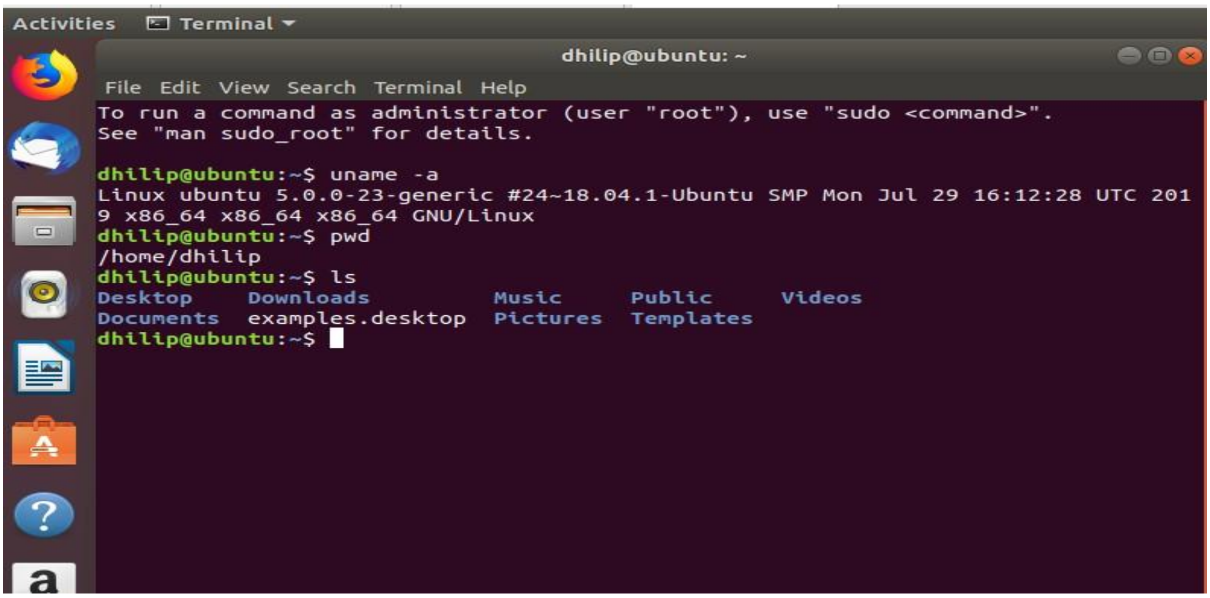


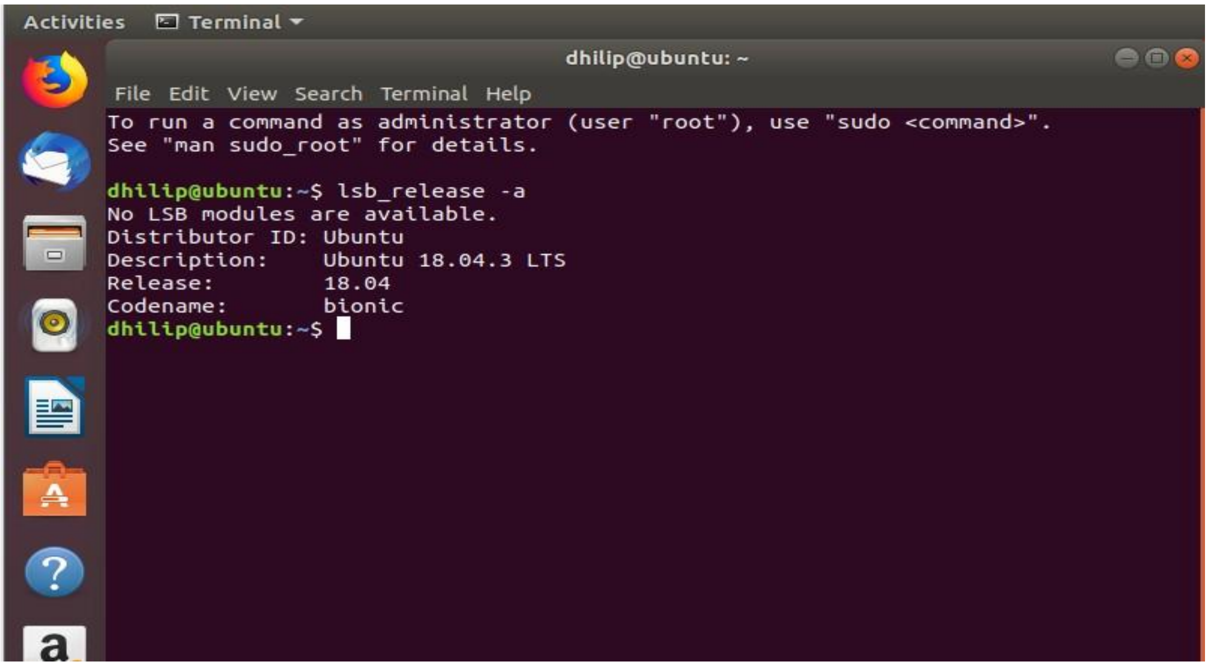





Ubuntu-VM
Sat 23:54
k.dhilip
Not listed?


Ubuntu-VM
Sat 23:55
k.dhilip
Password:
Cancel
Sign In


Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 201
9 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$ pwd
/home/dhilip
dhilip@ubuntu:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
dhilip@ubuntu:~$


Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 18.04.3 LTS
Release: 18.04
Codename: bionic
dhilip@ubuntu:~$

19. **Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.**

Aim:

To install a C compiler (GCC) in the Ubuntu virtual machine created using VMware Workstation and execute a simple C program.

Procedure:

Step 1: Start Ubuntu VM

1. Open VMware Workstation.
2. Start your Ubuntu VM.
3. Login to Ubuntu. Step 2: Open Terminal

Press:

Ctrl + Alt + T

Step 3: Update the Package List

Type:

sudo apt update

Press Enter.

Enter your Ubuntu password when asked.

While typing the password, you won't see any characters. This is normal. Type the password and press Enter.

Wait until the update finishes.

Step 4: Install GCC Compiler

Type: sudo apt

install gcc

If it asks:

Do you want to continue? [Y/n]

Type:

Y

and press Enter.

Wait for the installation to complete.

Step 5: Check GCC Installation

Type: gcc --

version

You should see something similar to: gcc

(Ubuntu ...) ...

This confirms that the C compiler is installed.

Step 6: Create a C Program

Create a file named hello.c:

nano hello.c The Nano

editor will open.

Type this program:

```
#include <stdio.h>

int main() {     printf("Hello
World!\n");     return 0; }
```

Step 7: Save the Program In

Nano:

1. Press Ctrl + O
2. Press Enter
3. Press Ctrl + X

You will return to the terminal. Step

8: Compile the C Program

Type:

gcc hello.c -o hello

If there is no error message, compilation was successful.

Step 9: Execute the Program

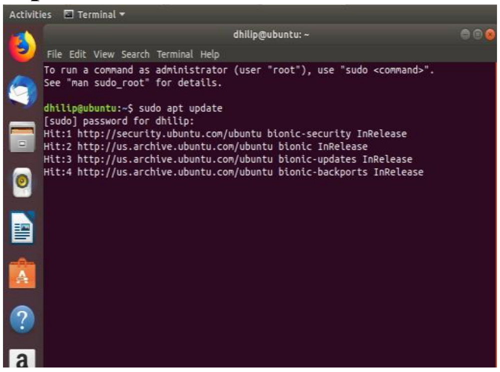
Type:

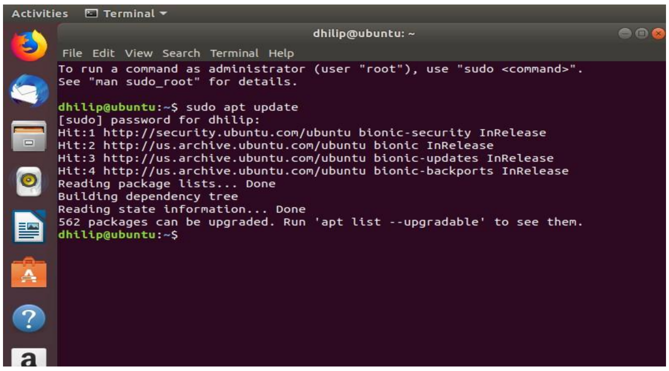
./hello

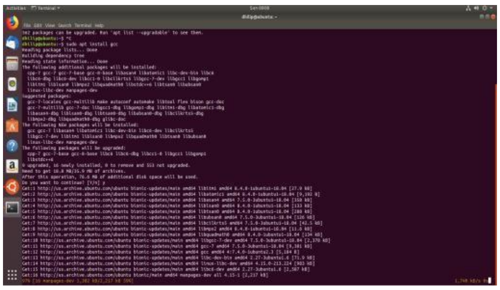
Expected output:

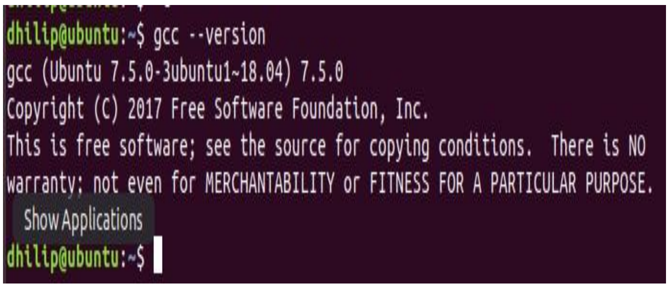
Hello World!

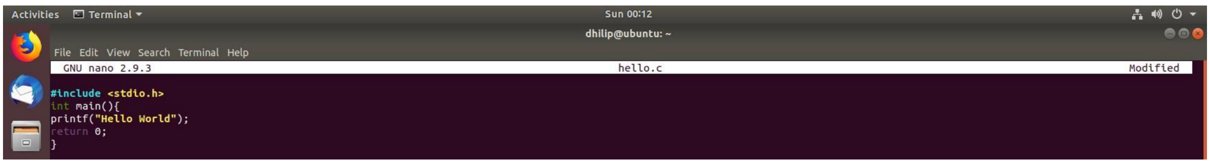
Implementation:











20. Install Virtualbox with different flavours of linux or windows OS on top of windows 10 using custom installation.

Aim:

To install Oracle VirtualBox on Windows 10 and install a different flavour of Linux/Windows operating system using Custom Installation.

Procedure:

Step 1: Open VirtualBox

1. Open Oracle VM VirtualBox on your Windows computer.
2. Click New to create a new virtual machine.

Step 2: Create the Virtual Machine

1. Enter the VM name, for example:

   Ubuntu-VirtualBox

2. Select the location where you want to store the VM.
3. Select the ISO Image of Ubuntu.
4. Select the operating system type:
   - Type: Linux ○
     Version: Ubuntu (64-bit)
5. Click Next.

Step 3: Select Custom Installation

When VirtualBox provides the installation options, choose the option for custom/unattended installation settings as applicable.

If you see Skip Unattended Installation, select it if you want to perform the OS installation manually.

Click Next.

Step 4: Configure Hardware

Set the hardware resources:

Memory: 2048

MB (2 GB)

Processors:

2 CPUs

Click Next.

Step 5: Create Virtual Hard Disk

1. Select Create a Virtual Hard Disk Now.
2. Set the disk size to:

20 GB

3. Click Next.
4. Review the configuration.
5. Click Finish.

Step 6: Start the Virtual Machine

1. Select Ubuntu-VirtualBox.
2. Click Start.
3. Ubuntu will boot from the selected ISO file.

Step 7: Install Ubuntu

When the Ubuntu installer appears:

1. Select Install Ubuntu.
2. Select the keyboard layout.
3. Choose the installation type.
4. Select Erase disk and install Ubuntu.

This refers to the virtual hard disk, not your actual Windows C: drive.

5. Set your location/time zone.
6. Create a username and password.
7. Click Install.
8. Wait for the installation to complete. Step 8: Restart the VM

After installation:

1. Click Restart Now.
2. If VirtualBox asks you to remove the installation medium, press Enter.
3. Ubuntu will restart.
4. Login using the username and password you created. Step 9: Verify the

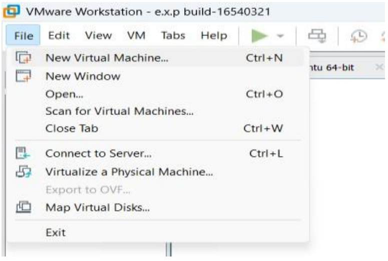
Installation

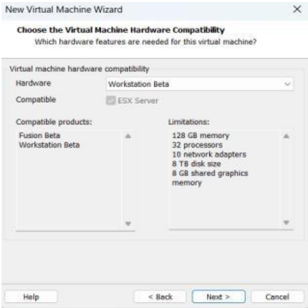
Open Terminal:

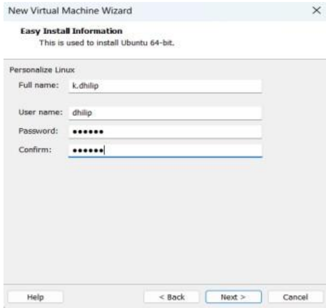
Run:

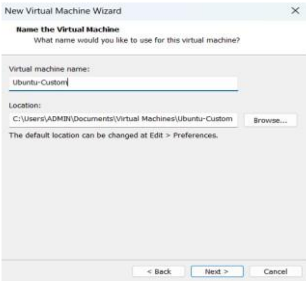
lsb_release –a

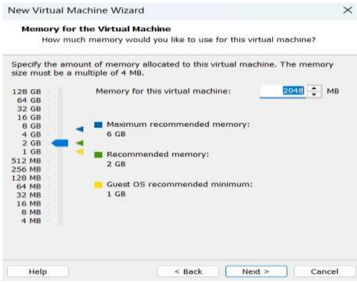
Implementation:

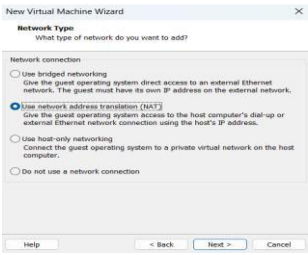


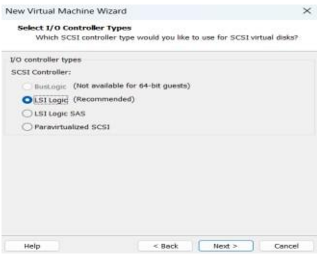


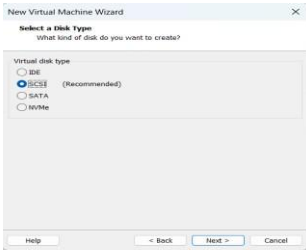


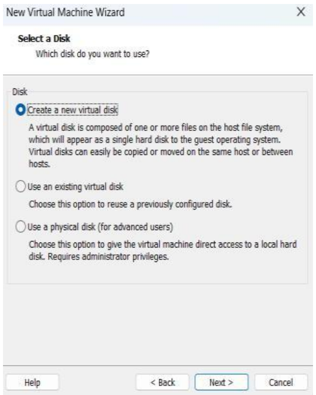


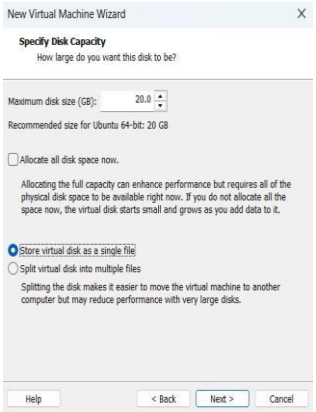


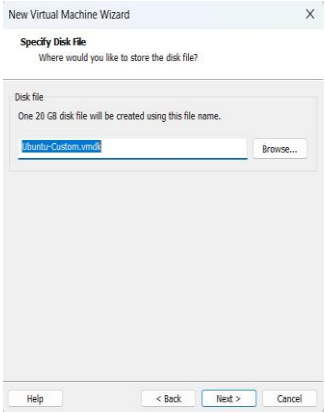


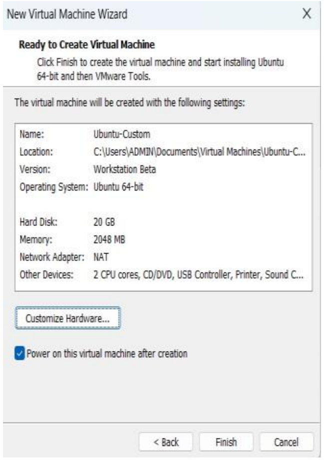












21. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows 7 or 8.

Aim:

To install VMware Workstation and create virtual machines with different operating systems such as Ubuntu Linux on the existing Windows environment.

Procedure:

Step 1: Open VMware Workstation

3. Open VMware Workstation in Windows.
4. Click Create a New Virtual Machine.

Step 2: Select Installation Method

3. Select Typical (recommended).
4. Click Next.

Step 3: Select the Operating System ISO

5. Select Installer disc image file (ISO).
6. Click Browse.
7. Select the Linux/Windows ISO file.
8. Click Next.

Example:

Ubuntu ISO → ubuntu.iso

Step 4: Select Operating System

3. Select the appropriate operating system.
4. For Ubuntu, select:

Linux Ubuntu
64-bit

4. Click Next.

Step 5: Give Virtual Machine Name

Enter a name such as:

Ubuntu-VM

Select a location where you want to store the VM.

Click Next.

Step 6: Configure Disk

2. Specify the disk size.

For example:

20 GB

4. Select Store virtual disk as a single file.
5. Click Next. Step 7: Customize Hardware

Click Customize Hardware.

Configure:

- Memory: 2 GB or more
- Processor: 2 cores
- Network Adapter: NAT

Click Close → Finish.

Step 8: Install the Operating System

6. Start the virtual machine by clicking Power on this virtual machine.
7. The operating system installer will start.
8. Follow the installation instructions.
9. Create a username and password when requested.
10. Wait for the installation to complete. Step 9: Restart the Virtual Machine

After installation:

4. Restart the VM.
5. Remove the ISO if VMware asks you to do so.
6. Ubuntu will boot from the virtual hard disk.

Step 10: Verify the Installation Open

the Terminal in Ubuntu and run:

uname -a

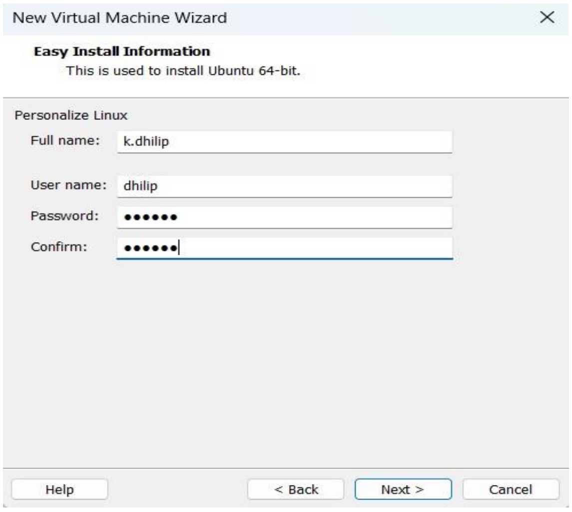
Then:

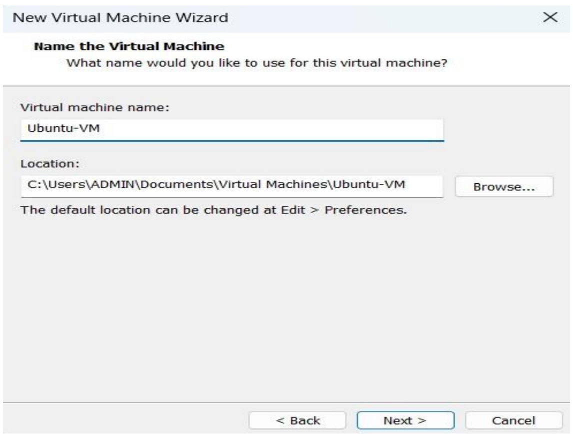
ls

And:

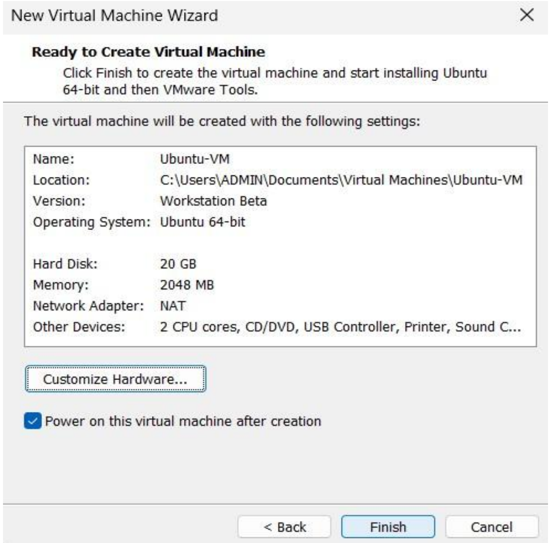
Pwd

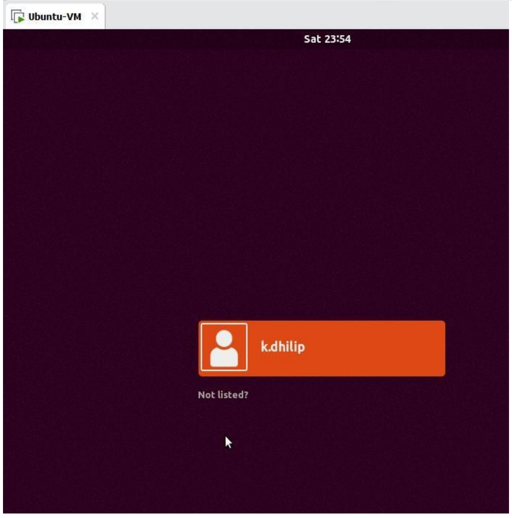
Implementation:

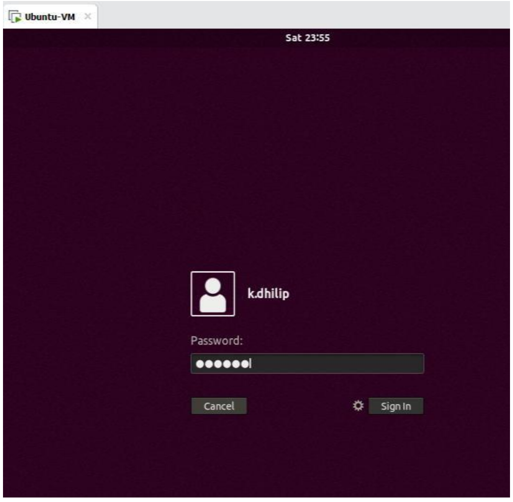


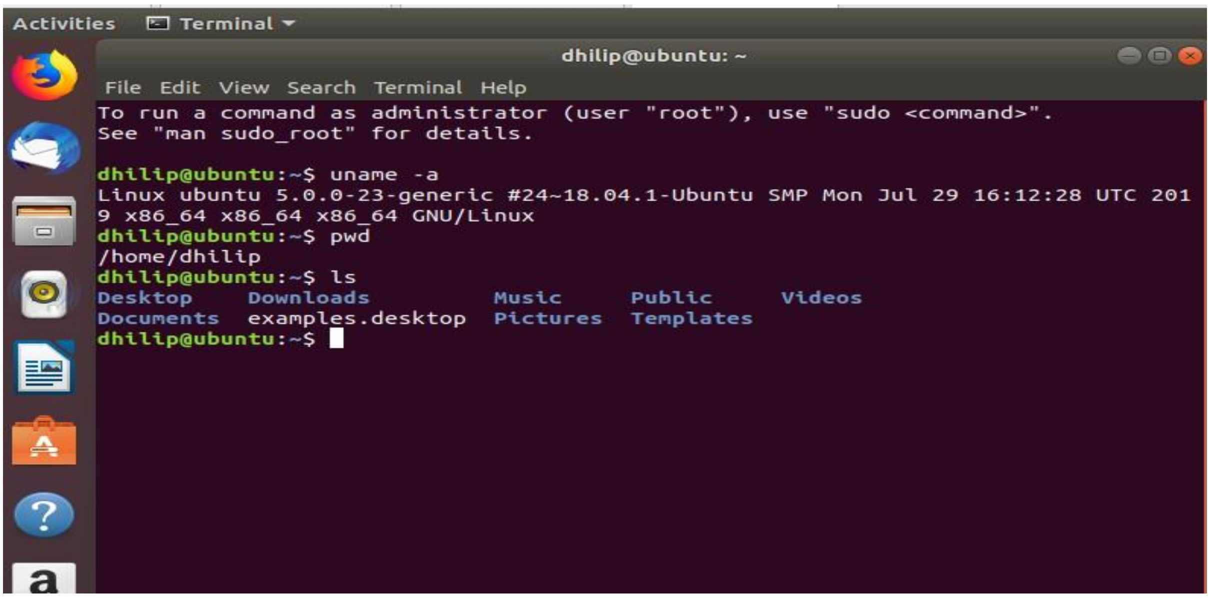


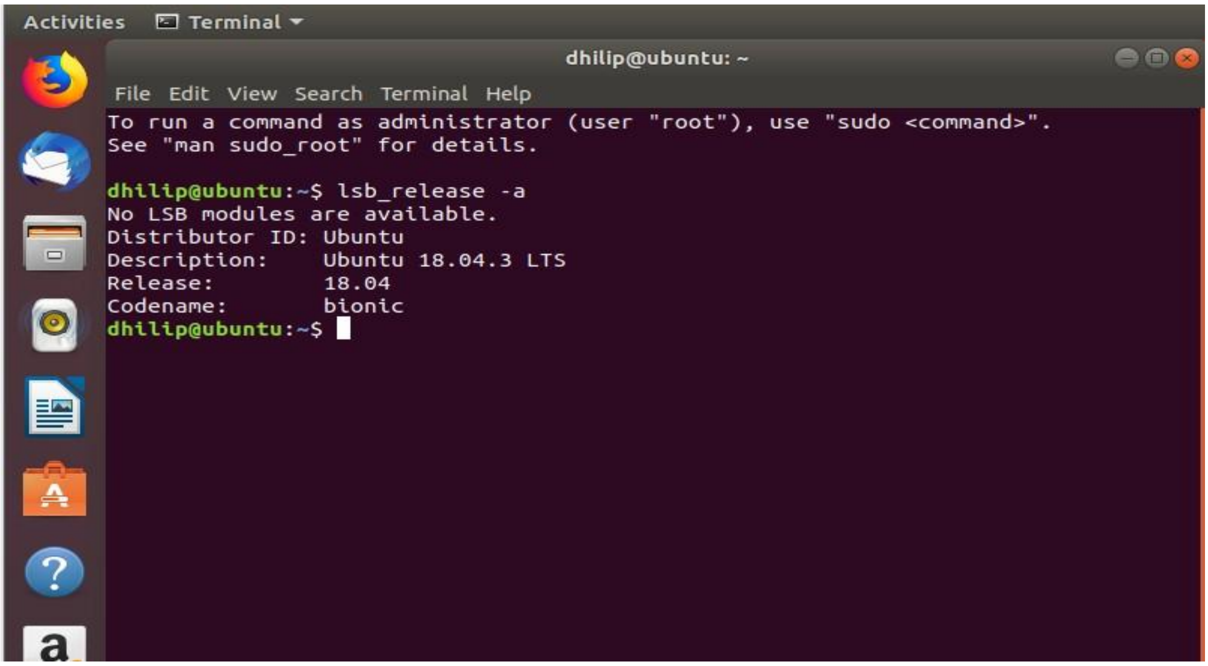

New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 20
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Ubuntu-VM
Location: C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM
Version: Workstation Beta
Operating System: Ubuntu 64-bit
Hard Disk: 20 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel


Ubuntu-VM
Sat 23:54
k.dhilip
Not listed?


Ubuntu-VM
Sat 23:55
k.dhilip
Password:
Cancel
Sign In


Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$ pwd
/home/dhilip
dhilip@ubuntu:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
dhilip@ubuntu:~$



22. To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

Aim:

To run virtual machines with different hardware configurations in VMware Workstation and determine how many virtual machines can be utilized simultaneously on the host computer.

Procedure:

Step 1 — Open VMware Workstation

1. Open VMware Workstation Pro.
2. Make sure your previously created Ubuntu VMs are available.

Step 2 — Check the Host Computer Resources

Before starting multiple VMs, check your Windows computer's:

- RAM
- CPU
- Available disk space

You can check RAM and CPU using:

Task Manager → Performance

Step 3 — Configure the First VM

For example, configure Ubuntu VM 1 with:

RAM: 2 GB CPU: 1
processor, 2 cores

Start the VM.

Step 4 — Configure the Second VM

Create/select another VM and configure it with:

RAM: 2 GB CPU: 1
processor, 2 cores

Start the second VM without shutting down the first VM.

Step 5 — Configure Additional VMs

If your computer has sufficient resources, start another VM.

For example:

VM 1 → 2 GB RAM
VM 2 → 2 GB RAM
VM 3 → 2 GB RAM

Do not start more VMs if the host system becomes very slow or runs out of memory.

Step 6 — Check Running Virtual Machines

In VMware Workstation, observe which VMs are powered on.

You can also check Windows:

Task Manager → Performance → Memory / CPU

Observe the resource usage while the VMs are running.

Step 7 — Test Each VM

Open the Terminal in each running Ubuntu VM and execute:

uname -a

and: pwd

If the commands execute successfully, the VMs are running correctly.

Step 8 — Determine the Maximum Number

Start the VMs one by one.

After starting each VM, check:

- CPU usage
- RAM usage
- System responsiveness
- Whether the VM operates normally

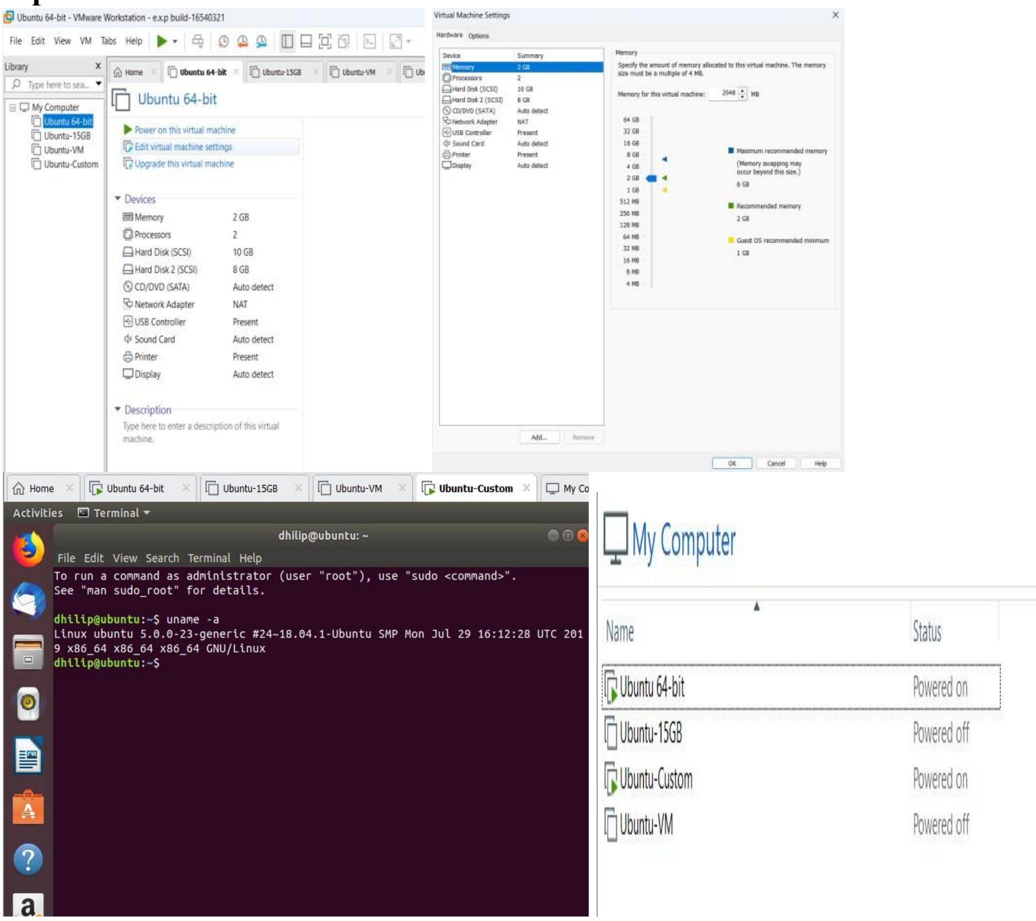
The maximum number of VMs is the number that can run simultaneously while the host and VMs remain usable.

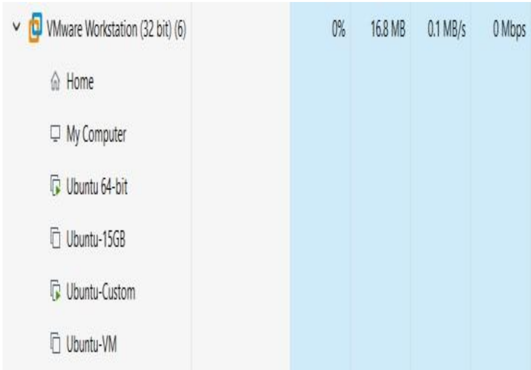
Example

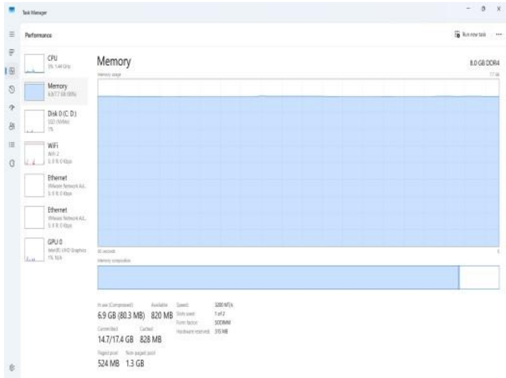
If your computer has 8 GB RAM, you might test:

VM 1 → 2 GB
VM 2 → 2 GB
VM 3 → 2 GB

Implementation:







23. Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

Aim:

To attach a virtual block (virtual hard disk) to a virtual machine and verify whether the data stored on it remains available even after the virtual machine is released or powered off.

Procedure:

We will use your Ubuntu VM in VMware Workstation Pro.

Step 1 — Power Off the VM

1. Open VMware Workstation Pro.
2. Select your Ubuntu VM.
3. Shut down Ubuntu completely.
4. Make sure the VM shows Powered Off.

Do not suspend it. It should be completely powered off.

Step 2 — Open Virtual Machine Settings

1. Select the Ubuntu VM.
2. Click Edit virtual machine settings.
3. The Virtual Machine Settings window will open.

Step 3 — Add a Virtual Hard Disk

1. Click Add...
2. Select:

Hard Disk

3. Click Next.

Step 4 — Select Disk Type

Select:

SCSI

Then click Next.

Step 5 — Create a New Virtual Disk

Select:

Create a new virtual disk

Click Next. Step 6 —

Set Disk Size

Enter:

1 GB

You can use 1 GB because we only need a small disk for this experiment.

Select:

Store virtual disk as a single file Click

Next.

Step 7 — Finish Adding the Disk Leave

the disk filename as the default.

Click:

Finish

Then click:

OK

The new virtual hard disk is now attached to your Ubuntu VM.

Step 8 — Start Ubuntu Power

on the Ubuntu VM.

Open Terminal:

Ctrl + Alt + T

Step 9 — Check the New Virtual Disk

Run: lsblk

You should see your original disk and a new disk, for example:

sda ├─sda1
└─sda2 sdb

The newly added disk will usually appear as sdb.

Do not format sda. That is your Ubuntu system disk.

Step 10 — Create a Partition If

the new disk is /dev/sdb, run:

sudo fdisk /dev/sdb

Inside fdisk:

Press:

n

Press Enter.

Then press Enter for the default partition options until the partition is created.

Finally press:

w

This saves the partition. Step

11 — Format the Partition

Run: sudo mkfs.ext4

/dev/sdb1

This creates an ext4 filesystem on the new virtual disk.

Step 12 — Create a Mount Directory

Run:

sudo mkdir /mnt/virtualdisk Step

13 — Mount the Virtual Disk

Run:

sudo mount /dev/sdb1 /mnt/virtualdisk

Check:

df -h

You should see /mnt/virtualdisk. Step 14 —

Create Data on the Virtual Block Create a

file:

sudo touch /mnt/virtualdisk/test_data.txt

Put some text into it: echo "Virtual Block Test Data" | sudo tee

/mnt/virtualdisk/test_data.txt

Check it:

cat /mnt/virtualdisk/test_data.txt

Expected:

Virtual Block Test Data

Data has now been stored on the additional virtual disk.

Step 15 — Release/Power Off the VM

1. Shut down Ubuntu:

sudo shutdown -h now

2. Wait until VMware shows the VM as Powered Off.

Step 16 — Start the VM Again Power

on the same Ubuntu VM.

Open Terminal.

Check the disk:

lsblk

You should see the additional disk again. Step

17 — Mount the Virtual Disk Again Run:

sudo mount /dev/sdb1 /mnt/virtualdisk

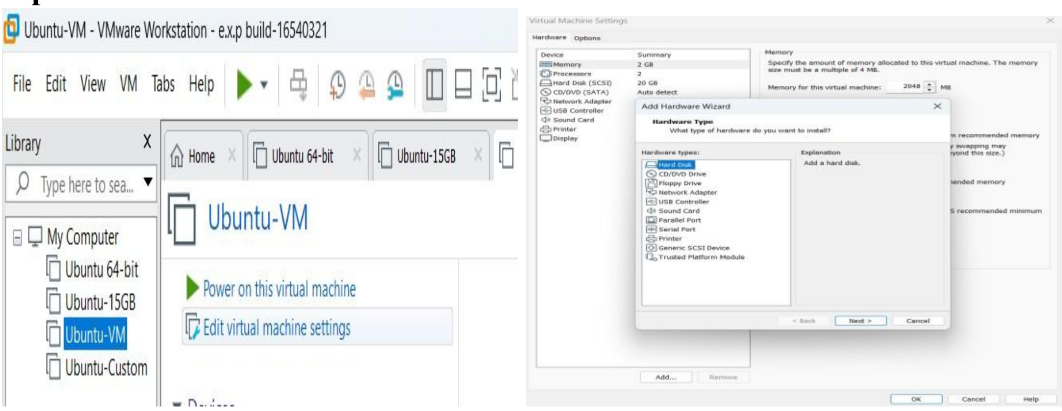
Then:

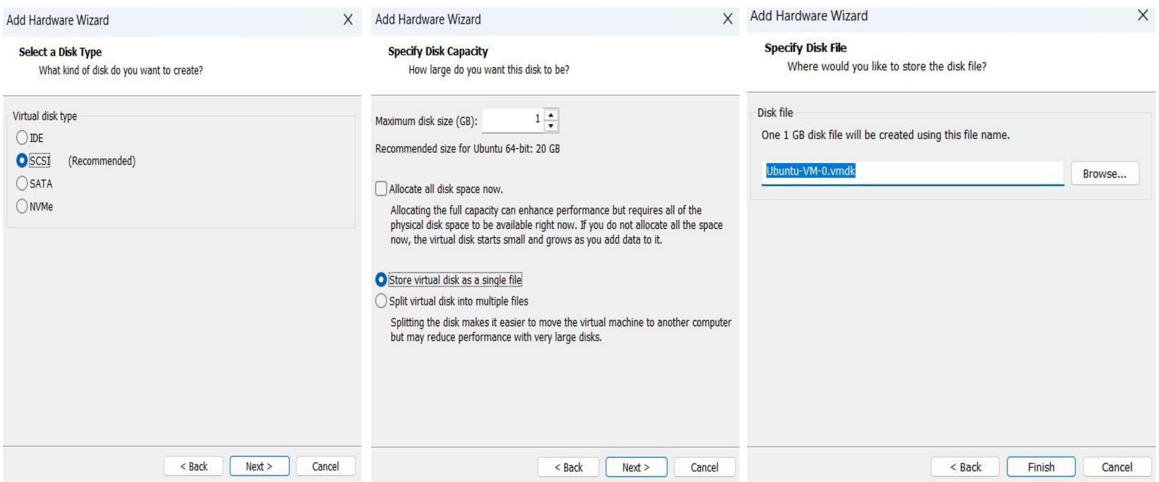
cat /mnt/virtualdisk/test_data.txt

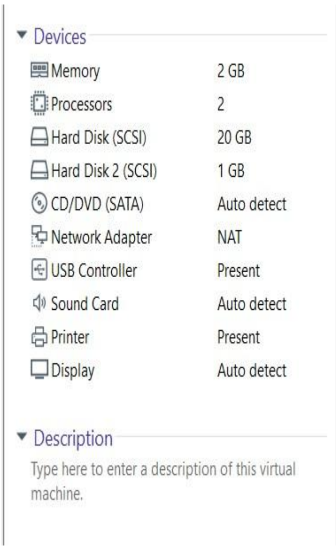
Expected Output

Virtual Block Test Data

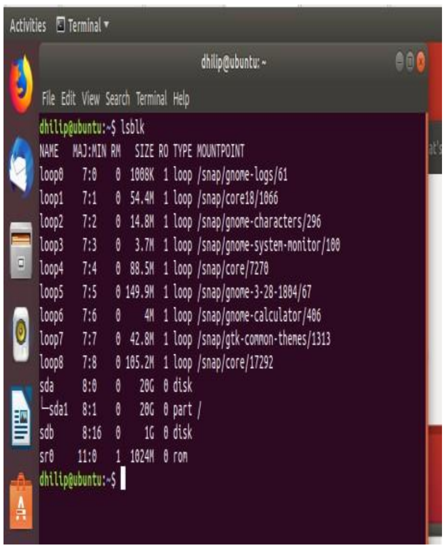
Implementation:

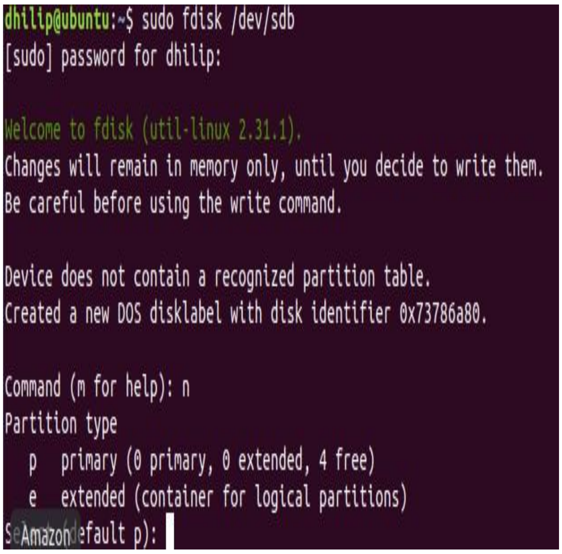


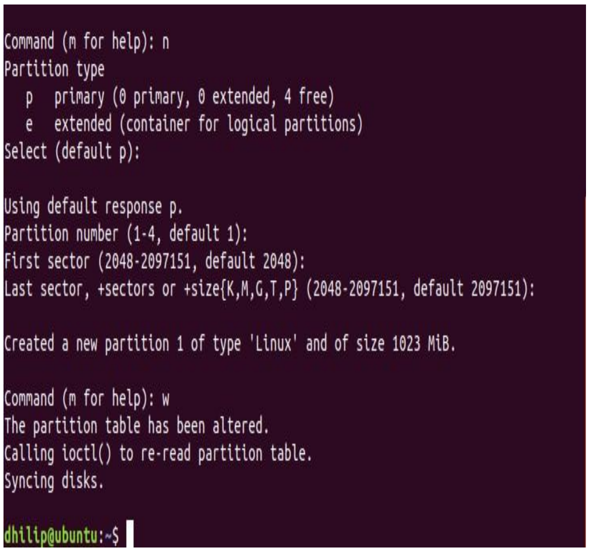



Devices
Memory 2 GB
Processors 2
Hard Disk (SCSI) 20 GB
Hard Disk 2 (SCSI) 1 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.


k.dhilip
Password:
Cancel
Sign In


Activities
Terminal
dhilip@ubuntu: ~
File Edit View Search Terminal Help
dhilip@ubuntu:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
loop0 7:0 0 1008K 1 loop /snap/gnome-logs/61
loop1 7:1 0 54.4M 1 loop /snap/core18/1066
loop2 7:2 0 14.8M 1 loop /snap/gnome-characters/296
loop3 7:3 0 3.7M 1 loop /snap/gnome-system-monitor/100
loop4 7:4 0 88.5M 1 loop /snap/core/7270
loop5 7:5 0 149.9M 1 loop /snap/gnome-3-28-1804/67
loop6 7:6 0 4M 1 loop /snap/gnome-calculator/406
loop7 7:7 0 42.8M 1 loop /snap/gtk-common-themes/1313
loop8 7:8 0 105.2M 1 loop /snap/core/17292
sda 8:0 0 20G 0 disk
└─sda1 8:1 0 20G 0 part /
sdb 8:16 0 1G 0 disk
sr0 11:0 1 1024M 0 rom
dhilip@ubuntu:~$


dhilip@ubuntu:~$ sudo fdisk /dev/sdb
[sudo] password for dhilip:
Welcome to fdisk (util-linux 2.31.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.
Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x73786a80.
Command (m for help): n
Partition type
p primary (0 primary, 0 extended, 4 free)
e extended (container for logical partitions)
Amazon
efault p):

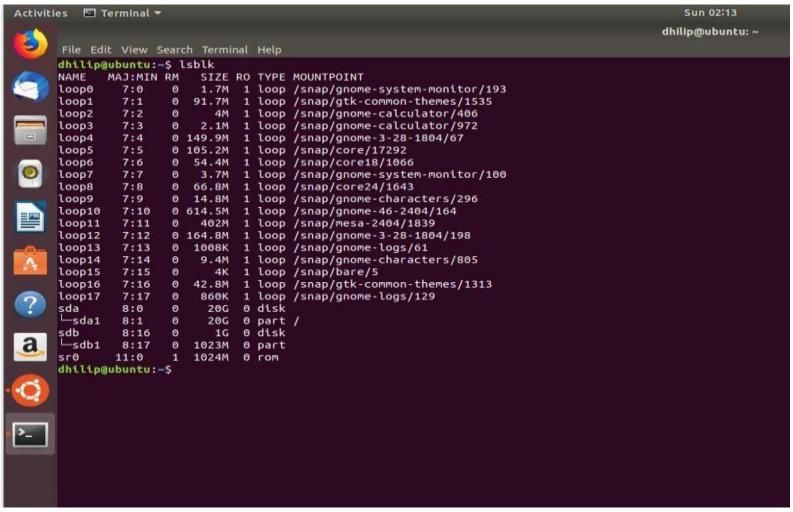

Command (m for help): n
Partition type
p primary (0 primary, 0 extended, 4 free)
e extended (container for logical partitions)
Select (default p):
Using default response p.
Partition number (1-4, default 1):
First sector (2048-2097151, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-2097151, default 2097151):
Created a new partition 1 of type 'Linux' and of size 1023 MiB.
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
dhilip@ubuntu:~$


dhilip@ubuntu:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.44.1 (24-Mar-2018)
Creating filesystem with 261888 4k blocks and 65536 inodes
Filesystem UUID: 9bdc26bb-5eaf-4499-9189-943da8ea1634
Superblock backups stored on blocks:
32768, 98304, 163840, 229376
Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done
dhilip@ubuntu:~$

```
dhilip@ubuntu:~$ sudo mkdir /mnt/virtualdisk
dhilip@ubuntu:~$ sudo mount /dev/sdb1 /mnt/virtualdisk
dhilip@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            960M     0  960M   0% /dev
tmpfs           197M  1.8M  195M   1% /run
/dev/sda1        20G  7.1G   12G  39% /
tmpfs           984M     0  984M   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
tmpfs           984M     0  984M   0% /sys/fs/cgroup
/dev/loop0      1.0M  1.0M     0 100% /snap/gnome-logs/61
/dev/loop1       55M   55M     0 100% /snap/core18/1066
/dev/loop2       15M   15M     0 100% /snap/gnome-characters/296
/dev/loop3      3.8M  3.8M     0 100% /snap/gnome-system-monitor/100
/dev/loop4       89M   89M     0 100% /snap/core/7270
/dev/loop5      150M  150M     0 100% /snap/gnome-3-28-1804/67
/dev/loop6      4.2M  4.2M     0 100% /snap/gnome-calculator/406
/dev/loop7       43M   43M     0 100% /snap/gtk-common-themes/1313
tmpfs           197M   16K  197M   1% /run/user/121
tmpfs           197M   24K  197M   1% /run/user/1000
/dev/loop8      106M  106M     0 100% /snap/core/17292
/dev/loop9       67M   67M     0 100% /snap/core24/1643
/dev/loop10     128K  128K     0 100% /snap/bare/5
/dev/loop11     896K  896K     0 100% /snap/gnome-logs/129
/dev/loop12     1.8M  1.8M     0 100% /snap/gnome-system-monitor/193
/dev/loop13     9.5M  9.5M     0 100% /snap/gnome-characters/805
/dev/loop14     2.2M  2.2M     0 100% /snap/gnome-calculator/972
/dev/loop15      92M   92M     0 100% /snap/gtk-common-themes/1535
/dev/loop16     165M  165M     0 100% /snap/gnome-3-28-1804/198
/dev/loop17     402M  402M     0 100% /snap/mesa-2404/1839
/dev/sdb1       991M  2.6M  922M   1% /mnt/virtualdisk
dhilip@ubuntu:~$
```

```
dhilip@ubuntu:~$ echo "Virtual Block Test Data" | sudo tee /mnt/virtualdisk/test__data.txt
Virtual Block Test Data
```

```
dhilip@ubuntu:~$ cat /mnt/virtualdisk/test__data.txt
Virtual Block Test Data
dhilip@ubuntu:~$
```



24. To showcase the virtual machine migration based on the certain condition from one host to the other.

Aim:

To demonstrate the migration of a virtual machine from one host/storage location to another using VMware Workstation Pro and verify the migrated virtual machine by executing basic Linux commands.

Procedure:

1. Power off the Ubuntu VM properly from Ubuntu.
2. Close VMware Workstation Pro.
3. Open File Explorer and navigate to:

   C:\Users\ADMIN\Documents\Virtual Machines

4. Locate the original VM folder:

   Ubuntu-VM

5. Copy the complete Ubuntu-VM folder.
6. Paste it in the same location and rename the copied folder as:

   Ubuntu-VM-Migrated

7. Open VMware Workstation Pro.
8. Select:
   File → Open
9. Navigate to:

   C:\Users\ADMIN\Documents\Virtual Machines\Ubuntu-VM-Migrated

10. Select the file with type:
    VMware virtual machine configuration (.vmx).
11. Click Open.
12. VMware displays:
    ―Did you move or copy this virtual machine?‖
13. Select:
    I Copied It

14. Click OK.

15. Power on the migrated Ubuntu virtual machine.
16. After Ubuntu starts, open the Terminal using:

Ctrl + Alt + T

17. Verify the migrated VM using:

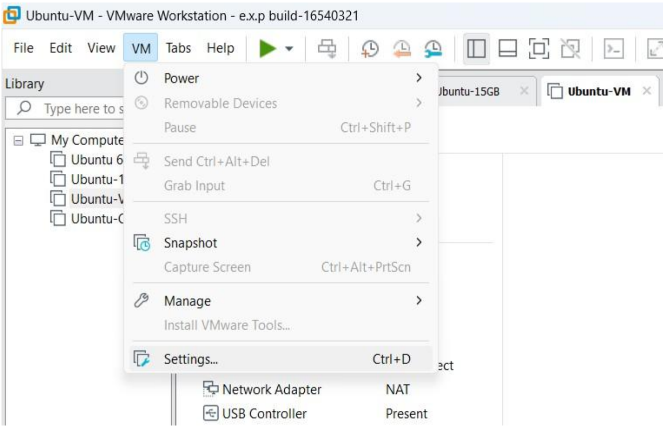
hostname

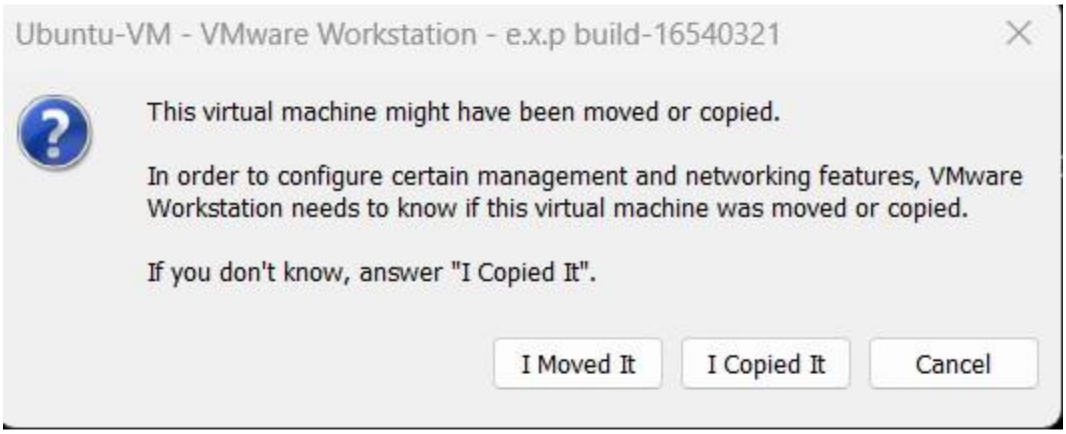
18. Check the Linux system information using:

uname -a

19. If the Ubuntu VM starts successfully and the commands execute correctly, the migration is successfully demonstrated.

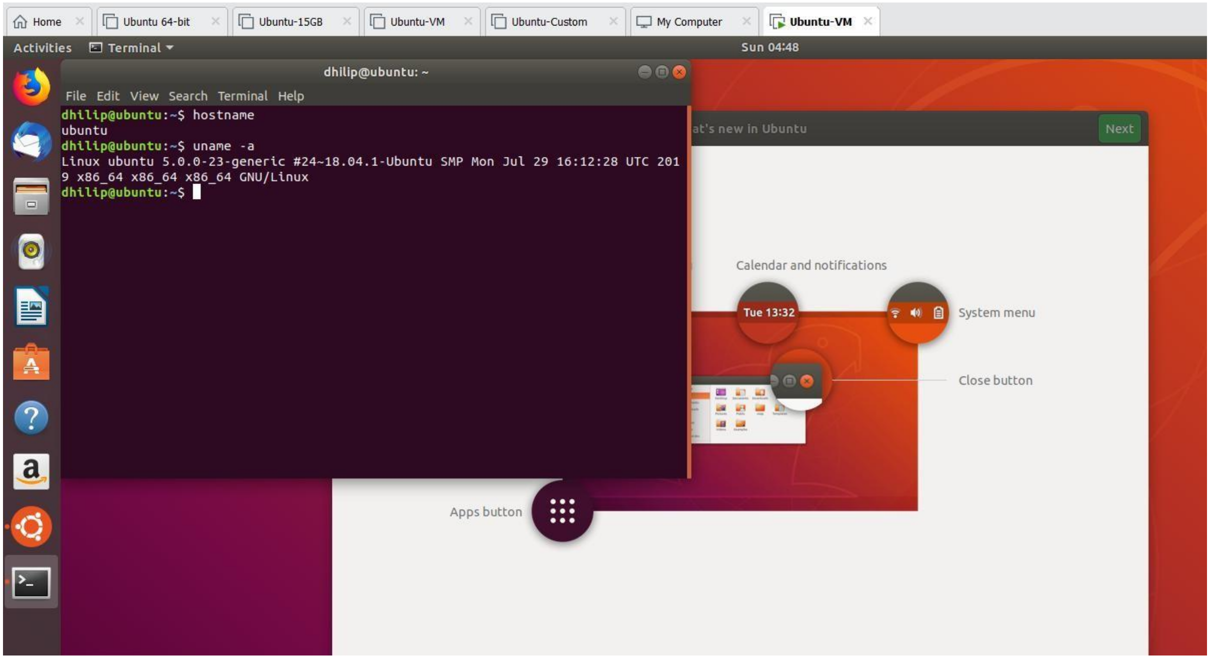
Implementation:






k.dhilip
Not listed?


k.dhilip
Password:
Cancel
Sign In


Home
Ubuntu 64-bit
Ubuntu-15GB
Ubuntu-VM
Ubuntu-Custom
My Computer
Ubuntu-VM
Activities
Terminal
Sun 04:48
dhilip@ubuntu: ~
File Edit View Search Terminal Help
dhilip@ubuntu:~$ hostname
ubuntu
dhilip@ubuntu:~$ uname -a
Linux ubuntu 5.0.0-23-generic #24~18.04.1-Ubuntu SMP Mon Jul 29 16:12:28 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux
dhilip@ubuntu:~$
at's new in Ubuntu
Next
Calendar and notifications
Tue 13:32
System menu
Close button
Apps button